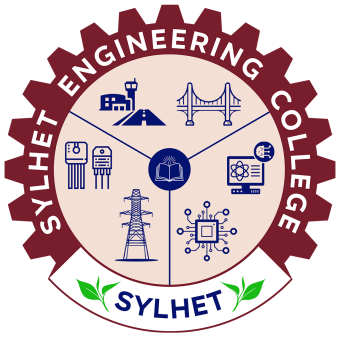
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Department of Computer Science & Engineering



Automatic Swara Identification in Continuous Indian Classical Flute Melodies With Comparative Analysis of Segmentation Techniques

Submitted by

Akash Datta Roni
Reg: 2020331550

Misbah Al Mamun
Reg: 2020331553

Department of Computer Science and Engineering
Sylhet Engineering College, Sylhet

Supervised by

Md. Abu Naser Majumder

Associate Professor and Head of the Department
Department of Computer Science and Engineering
Sylhet Engineering College, Sylhet

22 August 2026

Recommendation Letter from Thesis Supervisor

The thesis titled **"Automatic Swara Identification in Continuous Indian Classical Flute Melodies With Comparative Analysis of Segmentation Techniques,"** submitted by Akash Datta Roni and Misbah Al Mamun, is being presented under my supervision.

After thoroughly reviewing the thesis, I confirm my approval for its submission for examination.

Md. Abu Naser Majumder
Associate Professor and Head of the Department
Department of Computer Science and Engineering
Sylhet Engineering College, Sylhet
22 August 2026

Certificate of Acceptance

The thesis is titled **"Automatic Swara Identification in Continuous Indian Classical Flute Melodies With Comparative Analysis of Segmentation Techniques"** submitted by **Akash Datta Roni** and **Misbah Al Mamun**; Student ID: **2020331550** and **2020331553**; Session **2020-2021** to the Department of Computer Science and Engineering, Sylhet Engineering College, has been accepted as satisfactory in partial fulfilment of the requirements for the Degree of Bachelor of Science in Computer Science and Engineering.

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Abstract

Musical Information Retrieval (MIR) has emerged as an important research area in machine learning, with automatic note segmentation and recognition remaining challenging tasks. While considerable progress has been made for Western instruments such as the piano, comparatively little research has focused on Indian classical music and its instruments, resulting in limited datasets and methodologies for automatic swara transcription. This study proposes an end-to-end pipeline for automatic segmentation and swara classification of monophonic Indian classical flute melodies.

The proposed framework consists of two stages: note segmentation and swara classification. A dataset of 748 individual flute notes was collected through a combination of self-recordings and publicly available recordings. These notes were then used to generate a synthetic melody dataset that closely resembles continuous flute performances. An CRNN with an attention mechanism was employed to identify note boundaries within continuous melodies, while a CNN with an attention mechanism was trained to classify segmented notes using Constant-Q Transform (CQT) features.

To investigate the effectiveness of different acoustic representations, multiple feature extraction techniques, including Log-Mel Spectrogram, CQT, pitch, RMS energy, and their combinations, were evaluated for the segmentation stage. Experimental results demonstrate that CQT-based features achieved the best overall performance. The proposed segmentation model achieved a segmentation accuracy of **92.86%**, while the classification model achieved **97.80%** swara classification accuracy on correctly segmented melodies.

The results demonstrate that accurate end-to-end swara transcription can be achieved using a relatively small dataset when combined with synthetic data generation, careful model optimization, and appropriate feature representation. The proposed framework provides a practical and scalable approach for automatic transcription of monophonic Indian classical flute melodies and has the potential to be adapted to other monophonic musical instruments with minimal annotated data.

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Chapter 1

Introduction

Musical Information Retrieval (MIR) [1] has become an increasingly important research area due to the rapid growth of digital music platforms, machine learning, and artificial intelligence. Modern MIR systems enable computers to analyze, understand, and process musical content automatically, making them valuable in numerous applications, including automatic music transcription, music recommendation, source separation, instrument recognition, music retrieval, and audio tagging. As the demand for intelligent music analysis continues to increase, developing reliable methods for understanding musical performances has become a significant research challenge.

Musical instruments can generally be categorized as either monophonic or polyphonic based on the number of notes they can produce simultaneously. Polyphonic instruments, such as the piano, guitar, and harp, are capable of producing multiple notes at the same time, whereas monophonic instruments, including the flute, trumpet, clarinet, and saxophone, produce only a single note at any given moment. Although monophonic instruments eliminate the complexity of overlapping notes, accurately identifying note boundaries and classifying individual notes from continuous performances remains a challenging task due to variations in note duration, articulation, performer technique, and recording conditions.

Among the various research problems in MIR, Automatic Music Transcription (AMT) [2] has received considerable attention. The objective of AMT is to convert an audio performance into a symbolic representation, such as musical notes or notation, without manual intervention [3, 4]. While substantial progress has been achieved for widely used Western instruments, particularly the piano, comparatively fewer studies have focused on Indian classical instruments. Consequently, publicly available datasets, benchmark models, and standardized evaluation methodologies for Indian classical music remain limited. This lack of resources presents a significant challenge for researchers working on automatic transcription of Indian classical instruments.

One of the primary obstacles in developing machine learning models for musical analysis is the availability of annotated data. Deep learning models typically benefit from large, carefully labeled datasets, which are both time-consuming and expensive to create. In the context of Indian classical music, the challenge is even greater because expert knowledge is often required to accurately annotate individual swaras and note boundaries. Furthermore, high-quality recordings of isolated notes and continuous performances are relatively scarce compared to those available for more widely studied instruments. As a result, collecting sufficiently large datasets for supervised learning becomes a major limitation.

To address these challenges, this research investigates whether accurate automatic swara transcription can be achieved using a relatively small amount of training data. A dataset consisting of 748 individual flute note recordings was collected through self-recordings and publicly available audio sources. Using these recordings, a synthetic melody dataset containing 1,000 continuous flute melodies was generated to simulate realistic musical performances while providing accurate ground-truth annotations for segmentation and classification.

The proposed framework consists of two sequential stages: note segmentation and swara classification. First, a convolutional recurrent neural network (CRNN) [5] with an attention mechanism is employed to detect note boundaries in continuous flute melodies. The segmented notes are subsequently classified using a convolutional neural network (CNN) [6] with an attention mechanism [7]. Throughout this research, several acoustic feature representations, including Log-Mel Spectrograms [8], pitch, RMS energy, and the Constant-Q Transform (CQT) [9], were investigated to determine the most effective representation for the proposed system.

Experimental results demonstrate that CQT-based feature extraction consistently outperformed the alternative representations evaluated in this study. The proposed segmentation model achieved a segmentation accuracy of 92.86%, while the classification model achieved a swara classification accuracy of 97.80% on correctly segmented melodies. These findings indicate that accurate end-to-end swara transcription can be achieved using a relatively small dataset when combined with appropriate feature engineering, synthetic data generation, careful model optimization, and hyperparameter tuning.

The primary contribution of this research is the development of an end-to-end pipeline for automatic segmentation and swara classification of continuous Indian classical flute melodies. Unlike many existing approaches that rely on large annotated datasets, the proposed framework demonstrates that competitive performance can be achieved with limited training data by combining synthetic data generation with suitable feature representations and attention-based deep learning models. The proposed methodology also provides a practical foundation for extending automatic music transcription techniques to other monophonic musical instruments that suffer from limited annotated data.

1.1 Motivation

Automatic Music Transcription (AMT) has attracted significant attention in recent years due to its wide range of applications in music education, digital archiving, music retrieval, and intelligent music analysis. However, most existing research has primarily focused on Western musical instruments, particularly the piano, because of the availability of large-scale annotated datasets and established benchmark models. In contrast, Indian classical instruments, especially the flute, remain relatively underexplored despite their cultural significance and widespread use.

One of the major barriers to research in this area is the scarcity of high-quality labeled datasets.

Creating annotated musical data requires considerable time, expert knowledge, and manual effort to accurately identify note boundaries and swaras. Consequently, developing deep learning models for Indian classical music becomes difficult, as conventional approaches often rely on thousands of labeled recordings for effective training.

This research was motivated by the need to develop an efficient and practical automatic swara transcription system that can achieve high performance without requiring massive annotated datasets. Instead of depending solely on large collections of real recordings, this work investigates whether synthetic melody generation, combined with appropriate feature extraction techniques and attention-based deep learning models, can provide a reliable alternative. Furthermore, this study aims to contribute to the relatively limited body of research on Indian classical flute analysis and provide a foundation for future work on other monophonic musical instruments.

1.2 Objectives

The primary objective of this research is to develop an end-to-end deep learning framework capable of automatically segmenting and classifying swaras from continuous Indian classical flute melodies. To accomplish this objective, the following goals were established:

- To collect and prepare a dataset of isolated Indian classical flute notes.
- To generate a synthetic melody dataset that accurately represents continuous flute performances while providing precise ground-truth annotations.
- To develop an automatic note segmentation model capable of identifying note boundaries in continuous audio recordings.
- To investigate and compare multiple acoustic feature representations, including Log-Mel Spectrogram, pitch, RMS energy, and Constant-Q Transform (CQT), for note segmentation and swara classification.
- To design and evaluate attention-based deep learning models for both segmentation and classification tasks.
- To analyze the effectiveness of the proposed framework using quantitative performance metrics and transcription results.
- To investigate whether accurate swara transcription can be achieved using a relatively small amount of training data through careful feature engineering, synthetic data generation, and model optimization.

Chapter 2

Literature Review

2.1 Introduction

Automatic Music Transcription (AMT) has been studied for several decades, but the great majority of published work has concentrated on instruments and traditions for which large annotated datasets already exist. This chapter reviews the research that shaped the design decisions made in this thesis: general approaches to AMT, the particular difficulties associated with monophonic instruments and low-resource corpora, acoustic feature representations used for note-level analysis, deep learning architectures used for onset detection and classification, prior work on Indian classical music, and techniques for generating synthetic training data.

Read together, these strands motivate the two central choices made in this work: building a synthetic melody corpus from isolated flute recordings, and pairing an CRNN-based segmentation stage with a CQT-driven CNN classifier. The chapter is organised as follows. Section 2.2 reviews general AMT frameworks; Section 2.3 covers monophonic transcription and low-resource instrument corpora; Sections 2.4 and 2.5 examine feature representations and neural architectures respectively; Section 2.6 surveys work specific to Indian classical music; Section 2.7 addresses synthetic data generation; and Section 2.8 states the research gap this thesis addresses.

2.2 Automatic Music Transcription: General Approaches

The task of AMT, converting an audio recording into a symbolic representation of pitch, onset and duration, has been described by Benetos et al. [3] as one of the long-standing open problems in music information retrieval, owing to the range of sub-tasks it subsumes: pitch estimation, onset and offset detection, instrument identification and, for polyphonic material, source separation. A later overview by the same group [4] organises the field into frame-level, note-level, stream-level and notation-level transcription, a hierarchy that is useful here because the present work operates at the note level: it must recover boundaries and labels, not merely a frame-wise pitch curve. A recent systematic survey by Jamshidi et al. [10] traces how the field moved from non-negative matrix factorization methods toward neural approaches, and notes that convolutional and recurrent architectures now dominate published AMT systems because they can be trained end to end on spectrogram-like inputs without hand-engineered pitch-tracking rules.

Piano transcription has received the most attention within AMT, largely because datasets such as MAPS and MAESTRO provide thousands of hours of aligned audio-MIDI pairs. Hawthorne et al. [11] introduced the Onsets and Frames architecture, which pairs a convolutional-recurrent onset detector with a framewise pitch detector and conditions the latter on the former's predictions, an approach that substantially improved note F1-scores over prior systems on the MAPS benchmark. The core idea, that separating onset detection from sustained-pitch classification and then fusing the two produces cleaner note boundaries than either sub-task alone, carries over to the two-stage segmentation-then-classification design adopted in this thesis, even though the underlying instrument, dataset scale and network types differ substantially. The transfer is not automatic, however: the onset head in Onsets and Frames exploits the sharp percussive attack of a struck string, and the flute produces no such attack. Section 2.5 returns to this point.

2.3 Monophonic Transcription and Low-Resource Instrument Corpora

Before neural methods became standard, monophonic pitch estimation was handled by signal-processing algorithms that remain strong baselines. YIN [12] estimates the fundamental period from the autocorrelation-derived difference function in the time domain, while SWIPE [13] matches the spectrum against a sawtooth-inspired kernel in the frequency domain; pYIN [14] later added a hidden Markov layer over YIN candidates to produce smoother pitch tracks. These methods are accurate on clean sustained tones and require no training data, but they estimate pitch rather than notes: converting a frame-wise pitch curve into discrete note events still needs heuristic segmentation, which degrades under rapid articulation, ornamentation and expressive pitch movement. That gap between pitch tracking and note segmentation is one of the reasons this thesis treats segmentation as a learned task rather than a post-processing step.

Much of the AMT literature targets polyphonic material, where several simultaneous pitches have to be disentangled. Monophonic transcription is simpler in principle, since only one note sounds at a time, but it remains far from solved for instruments that lack large labeled corpora. Sofronievski and Gerazov [15] built Scorpiano, a signal-processing pipeline for monophonic piano transcription that combines onset detection, tempo estimation and pitch tracking, and reported accuracy competitive with commercial neural systems while requiring no training data at all, a useful reminder that classical DSP techniques remain a viable baseline when annotated audio is scarce. Thomé and Ahlbäck [16] took a different route, training a convolutional-recurrent (ConvLSTM) pitch detector without an explicit musical language model and reporting an F-measure of 83% on the 2007 MIREX multi-F0 development set for a woodwind ensemble recording containing bassoon, clarinet, flute, horn and oboe. Their system is polyphonic rather than monophonic, so it is cited here not as a monophonic baseline but as evidence that a learned, timbre-agnostic representation handles flute-family material without instrument-specific tuning. The same argument is made more directly by Bittner et al. [17], whose lightweight instrument-agnostic model is trained jointly for onsets, multipitch and note activations and is explicitly

motivated by low-resource, instrument-agnostic transcription; their results show that a single compact model built on a harmonic CQT front end generalises across instrument families at a small cost in accuracy relative to specialised systems.

Two studies deal directly with monophonic transcription of underrepresented instruments and stand as the closest precedents to the present work. Loresco et al. [18] built a deep learning pipeline to transcribe the diwdiw-as, a Filipino bamboo panpipe, using a dataset collected specifically for the study, and reported 96.11% pitch accuracy and 94.07% note-duration accuracy. Their result shows that competitive transcription accuracy is attainable for a folk monophonic instrument without access to a MAPS-scale corpus, provided the pipeline is built around the available data rather than borrowed wholesale from piano AMT. Jancovic, Köküer and Baptiste [19], working on ornamented Irish traditional flute recordings, used hidden Markov models over MFCC features, with separate models for notes and for single-note ornaments, and reported that the soft onsets and very short durations of ornaments remain a major source of transcription error even when the underlying pitch content is estimated well. This is the most directly relevant caution for the present study: the instrument family is the same, and the synthetic melody generator used here introduces comparable articulation variation through its transition styles and DSP augmentations. Taken together, these papers frame the problem this thesis addresses: transcription systems for culturally significant but data-poor monophonic instruments cannot simply reuse piano-scale training regimes and must be designed around the data constraints from the outset.

2.4 Acoustic Feature Representations

Feature choice has a direct effect on segmentation and classification accuracy, and the transcription literature contains repeated comparisons of MFCC, Mel-spectrogram and Constant-Q Transform (CQT) representations. The CQT, in the efficient invertible formulation of Schörkhuber and Klapuri [20], places frequency bins on a geometric grid, so that the spacing between adjacent semitones is constant across the whole range. Two consequences matter for note-level work. First, a change of pitch becomes a fixed translation along the frequency axis, which suits convolutional kernels that share weights across that axis; the same is not true of a linear or Mel-scaled spectrogram, where the distance between semitones shrinks in the upper register. Second, the constant-Q property gives finer frequency resolution in the low register, where the fundamental of a bamboo flute lies, at the cost of coarser time resolution there. MFCCs, by contrast, were designed for speech: the cepstral truncation that makes them compact for phonetic modelling deliberately smooths away the fine harmonic structure that carries pitch identity.

Empirical comparisons support this reasoning without making it universal. Cheuk et al. [21] evaluated several input representations for neural transcription systems under matched conditions and found that the choice of time-frequency representation changes transcription accuracy materially rather than marginally. Hernandez-Olivan et al. [22] compared a Mel-based Onsets

and Frames model against a CQT-based salience model across the instrument families of the Slakh2100 dataset, which includes a flute subset, and reported that the CQT-based model degrades far less on instruments whose onsets are weaker than those of piano or plucked strings; training on isolated instrument families improved results further. Within Indian classical music specifically, Hebbar and Jagtap [23] compared preprocessing choices against several deep architectures for Carnatic raga recognition, establishing a precedent for treating the feature representation as a controlled experimental variable rather than a fixed design decision. The counterexample is equally instructive: Wolf-Monheim [24], evaluating spectral and rhythm features for general-purpose audio classification, found Mel-scaled spectrograms and MFCCs to outperform CQT and CENS chromagrams. Taken together, these results indicate that the best representation is task-dependent, that CQT tends to win where the target is note-level pitch content on a musical scale, and that the question should therefore be settled experimentally for the flute rather than assumed. That is the approach adopted in Chapter 4 of this thesis.

2.5 Neural Architectures for Onset Detection and Note Segmentation

Onset detection is the sub-task most directly analogous to the segmentation stage of this thesis. Schlüter and Böck [25] showed that a CNN operating on short log-Mel spectrogram excerpts could outperform the hand-designed spectral-flux detectors that had been the state of the art for over a decade, while requiring far less manual signal-processing tuning. Their result is frequently cited as early evidence that, for onset detection specifically, learned convolutional features generalise better than engineered ones. Eyben et al. [26] had earlier shown that bidirectional LSTMs trained on auditory spectral features could detect onsets across a wide range of instrument timbres without retuning, establishing that recurrent modelling of temporal context, and not just local spectral shape, is useful for this task. Çakır et al. [27] combined both ideas in a convolutional-recurrent architecture for polyphonic sound event detection and, in follow-up work, extended it with a frame-level attention mechanism so the network could down-weight irrelevant or silent frames. This is the same architectural family, CNN feature extraction, RNN temporal modelling and attention-based frame weighting, that underlies the CRNN-with-attention segmentation model used in this thesis. Attention has since been applied inside AMT itself: Wu et al. [28] use self-attention to frame multi-instrument transcription as an instance segmentation problem, treating note events as objects to be delineated rather than frames to be labelled independently.

A second and less frequently cited strand is more directly applicable to the flute, because the flute produces pitched, non-percussive onsets with a slow amplitude rise rather than a sharp transient. Stasiak et al. [29] fused several onset detection functions through a multi-layer perceptron rather than thresholding any single function, and evaluated across seventeen instruments and ensembles. Mońko and Stasiak [30] then applied a CNN onset detector specifically to bowed string recordings, the canonical soft-onset case, and found that a network trained on mixed material

transferred well to solo recordings of a sustained-tone instrument. Mounir et al. [31] approached the same problem from the feature side, proposing the NINOS² spectral sparsity measure and showing that it outperforms logarithmic spectral flux precisely where conventional detectors are weakest: sustained-string instruments, repeated notes and vibrato passages. These three papers matter for the present work because they identify soft onsets, repeated notes and vibrato as the failure modes of onset-based segmentation, and all three conditions occur in the synthetic flute melodies used here, both through the slurred and crossfaded transitions and through the vibrato augmentation. None of the reviewed onset-detection papers, however, evaluate on Indian classical instruments or on a corpus built from synthetically concatenated single-note recordings, which is the specific setting this thesis targets.

2.6 Note and Raga Recognition in Indian Classical Music

Computational work on Indian classical music (ICM) has been reviewed in detail by Singh and Biswas [32], who note that the field has concentrated on a small set of tasks, raga recognition foremost among them, and that note-level analysis remains comparatively underdeveloped. A difficulty specific to this repertoire is tonic (śruti) normalisation. Unlike Western fixed-pitch instruments, the absolute frequency of a given swara depends on the tonic (Sa) chosen by the performer, so the same swara label maps to different absolute pitches across recordings. Bellur et al. [33] proposed pitch-histogram and group-delay methods for identifying the tonic, and Gulati et al. [34] later compared the main approaches on a common evaluation set; Singh et al. [35] took the alternative route of building the tonic into the representation itself through a shruti-based chromagram. Many raga-recognition systems avoid the problem entirely by working with tonic-independent chroma or pitch-interval features. The segmentation stage of this thesis instead operates on absolute CQT contours, because boundary detection depends on spectral change rather than on scale-degree identity; the consequence is that the classification stage inherits a dependence on the tonic of the source recordings, and tonic normalisation is treated in Chapter 6 as a limitation and a direction for future work rather than as a solved problem.

At the level of melodic modelling, early work by Dighe et al. [36] used chromagram-derived swara histograms with classical classifiers, showing that swara occurrence and transition statistics carry strong discriminative information even without deep models. More recent systems use neural sequence models: DeepSRGM [37] applies an LSTM-based recurrent network to Carnatic recordings for raga classification and ranking, PhonoNet [38] uses a multi-stage CNN pipeline for Hindustani raga identification, and Shah et al. [39] report high accuracy on a ten-raga subset of the CompMusic collection using CNNs over raw spectrograms. Singh and Arora [40] revisit the task with explainability in mind and show that the regions their model attends to correspond to short melodic phrases rather than isolated notes, a useful caution for any system, including the one proposed here, that treats swara boundaries as if they were as sharply defined as piano note onsets. Two studies by Nuttall et al. [41, 42] work at the note level directly, modelling how a given svara's acoustic realisation is shaped by its melodic context and by gamaka

ornamentation in Carnatic music, through deep clustering and embedding methods respectively. Their finding that a svara's spectral shape is not fixed but depends heavily on its neighbours is directly relevant to the classification stage of this thesis, since the synthetic melody generator introduces comparable context-dependent variation through its transition-style and DSP augmentation parameters.

The closest published system to the present work is that of Singh et al. [43], who apply a convolutional-recurrent network with a CTC objective to transcription of Indian classical music. Their use of CTC is significant: it removes the need for frame-aligned labels, but in exchange the model predicts a sequence of symbols without committing to explicit onset and offset times. The framework proposed in this thesis takes the opposite position, predicting frame-level boundaries directly, which is what makes note-duration and timing information recoverable. Neither their work nor the raga literature surveyed above addresses the bamboo flute specifically, and none of it evaluates frame-level segmentation of continuous melodies into individual swaras.

2.7 Addressing Data Scarcity through Synthetic Data Generation

Because annotated recordings of Indian classical flute performances are scarce, the synthetic melody generation strategy adopted in this thesis sits within a broader body of work on synthetic and recombined training data. The closest precedent within music transcription is the Slakh dataset [44], which renders MIDI transcriptions through virtual instruments to obtain large volumes of audio with exact, automatically derived annotations, and which has since become a standard testbed for multi-instrument transcription. Slakh establishes the central argument for procedurally generated audio in this field: when ground truth is produced by the generator rather than by a human annotator, it is exact by construction and available at any scale. Its principal weakness is equally well known, in that the timbre of a sampled virtual instrument is not the timbre of a live performance.

The approach used here inverts that trade-off. Rather than synthesising audio from symbolic input, isolated real flute recordings are concatenated into continuous melodies with programmatically controlled transitions and DSP augmentations, so that every acoustic unit in the corpus is a real recording of the target instrument while the boundary annotations remain exact. Work elsewhere in the audio community follows the generative rather than the recombinative paradigm: Ghosh et al. [45] augment small audio classification datasets with a text-to-audio diffusion model whose outputs are aligned to the target distribution through preference optimisation, and Ibaraki and Chiang [46] generate training data for low-resource speech recognition by pairing novel text with text-to-speech synthesis. Both report that carefully constructed synthetic data substitutes for scarce real recordings rather than merely padding dataset size, which supports the general premise, but neither preserves the acoustic identity of a specific physical instrument in the way that recombination does. The limitation of the recombinative approach should be stated plainly, since it bears on how the results of this thesis are interpreted: concatenated melodies

reproduce the timbre of the source notes but not the continuous breath control, phrase-level dynamics and gamaka-like pitch movement of a live performance, so accuracy measured on this corpus is an upper bound on what would be obtained on unconstrained recordings. Chapter 6 discusses this gap and the evaluation on held-out real audio used to bound it.

2.8 Summary and Research Gap

Across this body of work, three patterns stand out. First, deep learning based AMT is well developed for piano and, increasingly, for polyphonic multi-instrument settings, but monophonic transcription for non-Western or folk instruments remains scattered across a handful of isolated studies, most of which build small custom datasets in a manner similar to the approach taken here. Second, CNN, RNN and attention architectures originally developed for onset detection and polyphonic sound event detection have been applied to soft-onset instruments such as bowed strings, but not, to the best of our knowledge, to frame-level segmentation of continuous Indian classical flute melodies into swara boundaries, even though the underlying ideas of frame-level attention, CQT input and CNN-RNN fusion transfer naturally to this setting. Third, the Indian classical music literature has concentrated on raga-level classification and, more recently, on context-dependent swara realization and CTC-based sequence transcription, but has not combined a segmentation-first pipeline with a systematic feature comparison and an evaluation on synthetically generated continuous melodies.

The framework developed in the remainder of this thesis sits at the intersection of these three gaps. Specifically, it makes three contributions:

(i) it adapts attention-based convolutional-recurrent segmentation, established in the general AMT and sound event detection literature, to the understudied domain of the monophonic Indian classical flute, where onsets are soft and note boundaries are not marked by percussive transients;

(ii) it evaluates CQT systematically against Mel-spectrogram, MFCC, pitch and RMS energy features for this specific timbre and task, rather than adopting a representation by convention; and

(iii) it introduces a recombination-based synthetic melody generator that removes the manual annotation bottleneck while keeping every acoustic unit grounded in a real flute recording, and that is instrument-independent by construction.

Chapter 3

Dataset Preparation

3.1 Dataset Preparation

Data Source. The audio recordings used for training the proposed model were collected from two primary sources: publicly available recordings from YouTube and recordings performed by the authors. The YouTube recordings were selected to obtain high-quality, low-noise samples, while the self-recorded flute performances were included to introduce realistic environmental noise and recording variations. Combining these two sources resulted in a more diverse dataset, enabling the model to learn from both clean and noisy audio conditions and thereby improving its robustness.

After collection, each audio clip was manually categorized according to its corresponding swara (Sa, Re, Ga, Ma, Pa, Dha, and Ni). The recordings were then organized into separate folders, with each folder containing audio samples of a single swara. This folder-based organization simplified data management and facilitated the preprocessing and training stages of the proposed framework. The directory structure of the dataset is illustrated in Figure 3.1.

```
recording for paper/
    re/
        re_53.wav
        re_42.wav
        ... (105 more)
    ga/
        ga_38.wav
        ga_35.wav
        ... (104 more)
    sa/
        sa_35.wav
        sa_89.wav
        ... (104 more)
    dha/
        dha_32.wav
        dha_53.wav
        ... (107 more)
    ni/
        ni_22.wav
        ni_34.wav
        ... (105 more)
    pa/
        pa_56.wav
        pa_65.wav
        ... (104 more)
    ma/
        ma_100.wav
        ma_10.wav
        ... (105 more)
```

Figure 3.1: Folder structure of dataset

Each swara class contains more than 100 audio recordings, providing a sufficiently balanced dataset for model training. Furthermore, the recordings exhibit a range of note durations, allowing the dataset to capture natural temporal variations in flute performances. The class distribution and the duration distribution of the individual swara recordings are presented in the following figures, Figure 3.2.

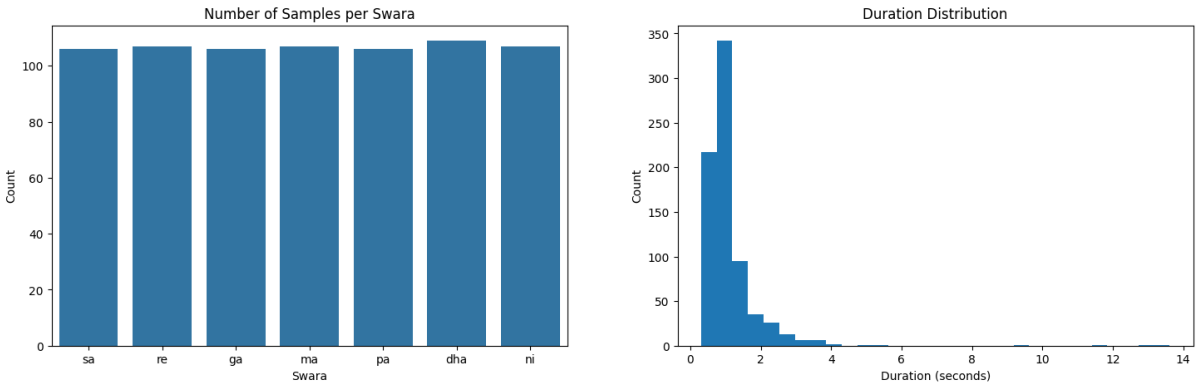


Figure 3.2: Class Distribution and Duration Distribution of Collected Swaras

Table 3.1: Individual Swara Dataset

Metric	Value
Total Recordings	748
Average Recording per Class	106.86
Number of Swara Class	7
Average Duration	1.137 (sec)
Median Duration	0.968 (sec)
Minimum Duration	0.305 (sec)
Maximum Duration	13.62
Standard Deviation of Duration	1.014 (sec)

Synthetic Melody Dataset Creation. To facilitate the evaluation of the proposed swara segmentation and classification framework, a synthetic melody dataset was generated by sequentially combining isolated flute recordings of the seven fundamental swaras (Sa, Re, Ga, Ma, Pa, Dha, and Ni), following the general recombination strategy used elsewhere to build large annotated corpora from isolated source material [44]. Each melody was constructed as an ordered sequence of individual note recordings, with every note assigned a unique sequence index to preserve its temporal position within the melody. During the generation process, the onset and offset timestamps of each swara were recorded, producing frame-level ground-truth annotations that precisely describe the temporal boundaries of every note in the synthesized melody. These annotations were stored in a metadata file containing the melody filename, note sequence index, swara label, start time, end time, transition styles and dsp applied. A portion of the melody dataset is presented in Table 3.2.

Table 3.2: Synthetic melody dataset

Melody file	Note_sequence	Swara_label	Start_time	End_time	Transition_style	Applied_dsp
flute_melody_0.wav	0	Sa	0	0.4043	phrase_start	speed(1.05)+vibrato+gain(1.02)
flute_melody_0.wav	1	Ma	0.3325	0.9921	slurred_crossfade	pitch(0.17)+speed(1.16)+vibrato+gain(0.63)+noise
flute_melody_0.wav	2	Pa	1.2282	3.4976	articulated_with_breath	pitch(1.06)+speed(0.82)+gain(1.23)+noise
flute_melody_0.wav	3	Ni	3.6154	4.678	articulated_with_breath	pitch(1.38)+speed(1.08)+gain(0.82)+noise
flute_melody_0.wav	4	Ni	4.9613	6.0312	articulated_with_breath	speed(0.99)+gain(1.22)

The synthetic melodies were generated by randomly selecting note recordings from the pool of isolated swara samples. For each melody, the sequence of swaras, the number of notes, and the specific recording chosen for each swara were determined programmatically to maximize diversity while ensuring that all seven swaras were represented throughout the dataset. Since multiple recordings were available for each swara, the generator selected different source recordings across melodies, preventing excessive repetition of identical note samples. Furthermore, the duration of each note within a melody was varied by assigning different temporal lengths, resulting in melodies with diverse rhythmic structures and overall durations. This randomization strategy produced a wide range of melodic patterns while preserving accurate ground-truth annotations for every note. The diverseness of notes per melody and duration is shown in Figure 3.3.

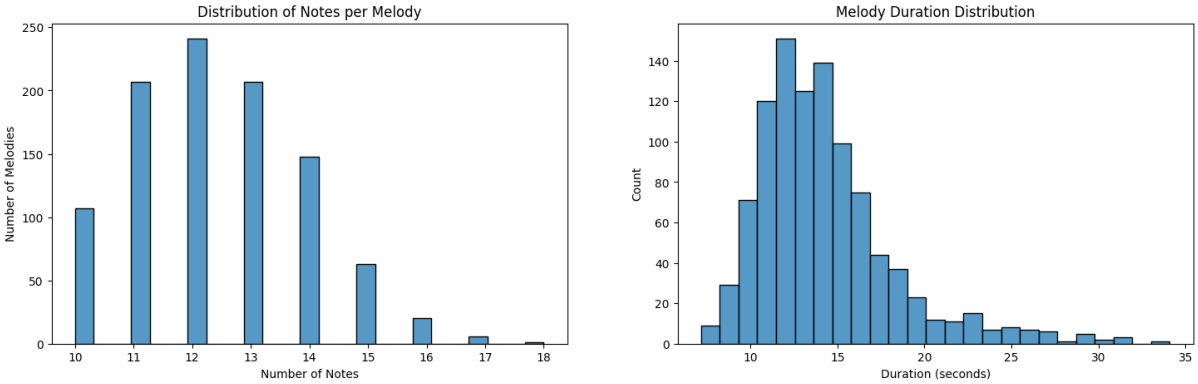


Figure 3.3: Melody Duration and Swaras used per Synthetic melody

To further improve the realism of the synthesized melodies, multiple transition techniques and digital signal processing (DSP) augmentations were incorporated during the generation process, in the same spirit as other DSP- and synthesis-based audio augmentation strategies used to compensate for scarce training data [45, 46]. The metadata indicates that transition styles such as direct transitions, crossfades, and silence-separated note boundaries were applied between consecutive swaras to simulate different articulation patterns observed in flute performances. In addition, several DSP-based augmentations, including pitch shifting, time stretching, gain adjustment, additive noise, vibrato, and reverberation, were selectively applied to individual notes or melody segments. The specific augmentation applied to each note was recorded in the metadata, allowing complete traceability of the synthesis process. These variations increase the acoustic diversity of the dataset and improve its suitability for evaluating the robustness and generalization capability of the proposed swara segmentation and classification framework. Figure 3.4 visualizes the usages of different DSP augmentation.

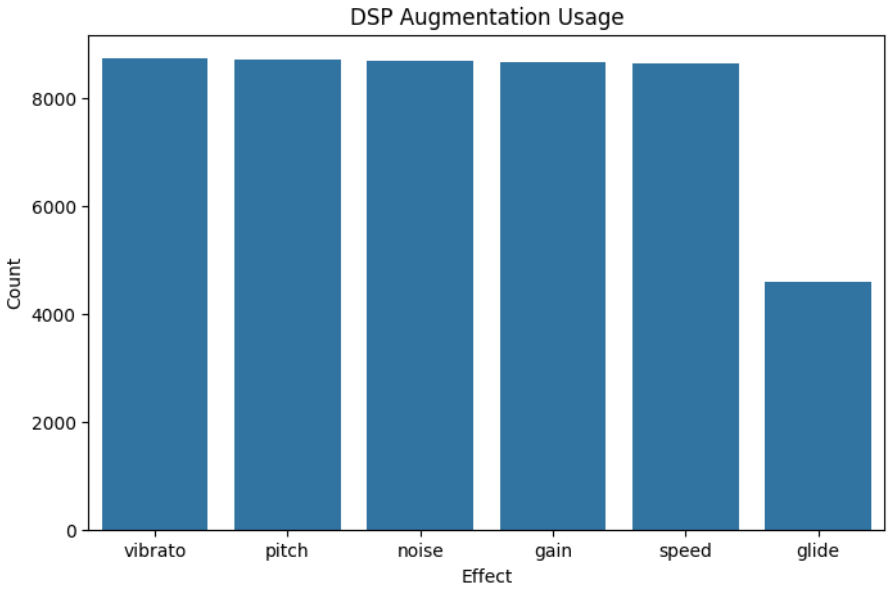


Figure 3.4: Count of different DSP augmentations used while creating Synthetic melody Dataset

The resulting metadata file serves as a comprehensive annotation source for the synthetic dataset, allowing direct computation of melody-level statistics, note durations, swara occurrence frequencies, transition patterns, and temporal distributions. These detailed annotations provide reliable ground truth for quantitative evaluation of both segmentation accuracy and swara classification performance.

The synthetic melody dataset was developed to address one of the primary challenges in automatic swara segmentation: the scarcity of accurately annotated continuous musical recordings. Creating a large-scale annotated dataset of flute performances requires substantial time and expert effort, as every note boundary and swara label must be manually identified and verified. Such a process is not only labor-intensive but also difficult to scale for different instruments and musical styles. To overcome this limitation, a synthetic melody generation pipeline was designed to automatically produce continuous melodies with precise ground-truth annotations, eliminating the need for extensive manual labeling.

Although the proposed framework is evaluated using Indian classical flute recordings, the underlying dataset generation pipeline is designed to be instrument-independent and can be adapted to any monophonic instrument for which isolated note recordings are available. By automatically generating continuous melodies from individual note samples and simultaneously producing accurate temporal annotations, the proposed approach significantly reduces the effort required to prepare training data. This enables the development of robust swara segmentation and classification models without relying on hundreds of hours of manually annotated recordings, making the framework more scalable and practical for future research on monophonic musical instruments.

Chapter 4

Methodology

4.1 Overview

This chapter describes the proposed end-to-end framework for automatic swara transcription from continuous Indian classical flute melodies. The proposed system consists of two major stages: note segmentation and swara classification. The segmentation model first identifies the boundaries of individual notes from a continuous melody. The segmented notes are then passed to the classification model, which predicts the corresponding swara for each note. Finally, the outputs of both models are combined to generate the complete swara transcription of the input melody.

Our general pipeline consists of the following phases:

1. Data Collection
2. Synthetic Melody Data Creation
3. Segmentation model (CRNN + Attention)
    a. Extract Melody features
    b. Use synthetic data and extracted features to train the model
    c. Evaluate segmentation success in actual melodies
4. Classification Model (CNN + Attention)
    a. Extract Individual Swara features
    b. Use extracted features to train the classification model
    c. Evaluation
5. Merge Both Models
6. Upload test melodies and evaluate

The proposed framework was divided into two sequential stages: note segmentation and swara classification. This design was adopted because the two tasks solve fundamentally different problems. The segmentation model is responsible for identifying the temporal boundaries of

individual notes within a continuous melody, whereas the classification model predicts the corresponding swara of each segmented note. Since the classification model was trained using isolated swara recordings, it requires individual note segments as input rather than continuous melodies. Therefore, accurate segmentation is an essential prerequisite for reliable swara classification. Furthermore, separating the pipeline enables independent evaluation of both stages, allowing segmentation and classification performance to be analyzed individually while simplifying model development, debugging, and future improvements.

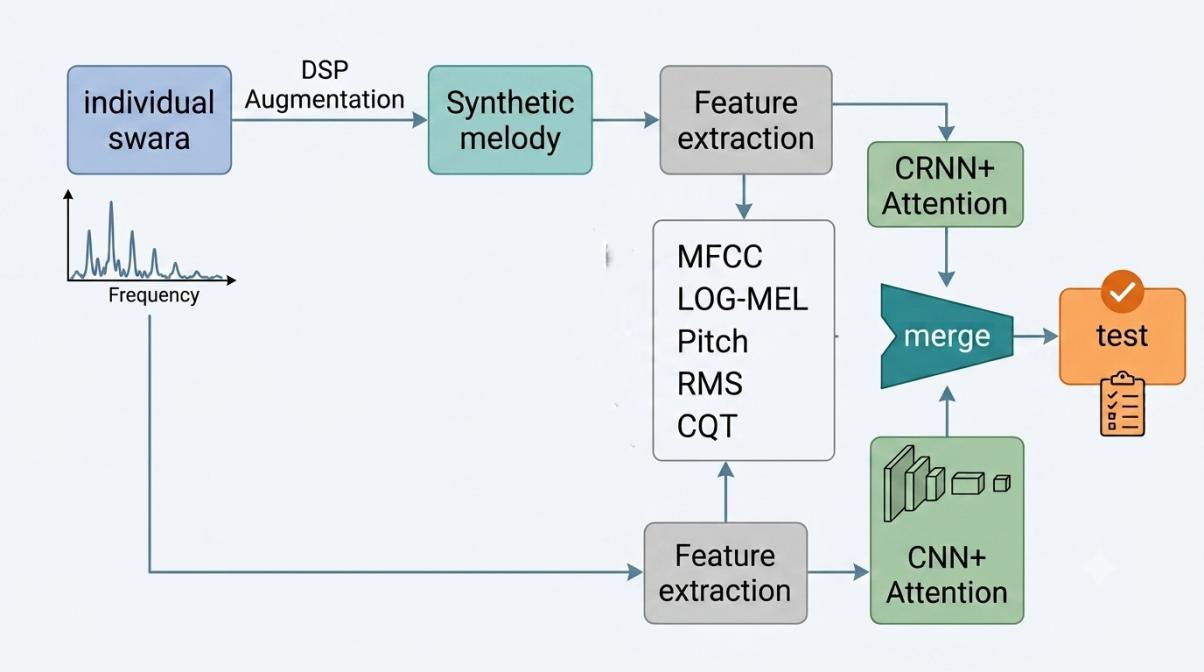


Figure 4.1: Methodology Overview

4.2 Overall Pipeline

4.2.1 Data collection

The first stage of the proposed framework involves collecting isolated recordings of the seven fundamental swaras of the Indian classical music scale: Sa, Re, Ga, Ma, Pa, Dha, and Ni. A total of 748 individual swara recordings were collected from both publicly available sources and self-recorded flute performances. Each recording was manually organized into separate folders according to its corresponding swara class. Each recording and were later resampled to be 16kHz, mono-channel and in .wav format.

These isolated recordings serve as the foundation of the proposed pipeline. They are used for two primary purposes: (i) generating the synthetic melody dataset required for training the segmentation model, and (ii) training the swara classification model to recognize individual notes. A detailed description of the data collection procedure, preprocessing steps, and dataset

statistics is provided in Chapter 3.

Data Splitting. For classification model, the isolated swara dataset was split into 80-10-10 for train test validation and the similar was done with synthetic melodies for segmentation training. Feature extraction was conducted and was split according to train-test-validation split.

4.2.2 Feature Extraction

For both segmentation and classification training, we extracted various features from the audio files, such as

I. **MFCC:** A compact representation of the spectral characteristics of audio that is widely used in speech and audio recognition tasks [47].

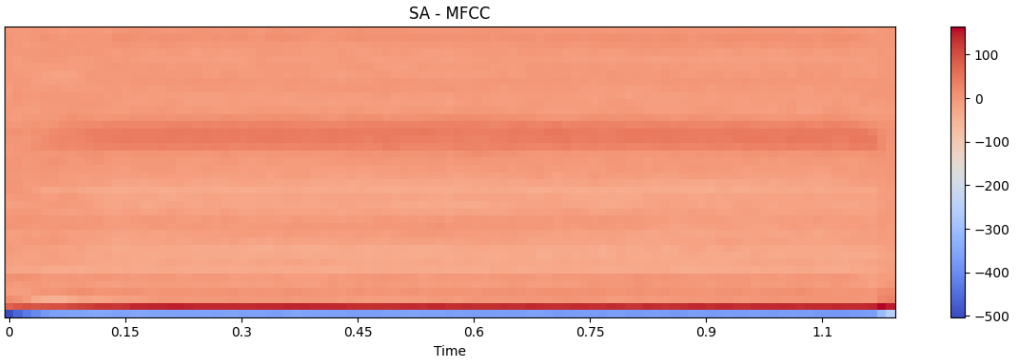


Figure 4.2: MFCC visualization of an individual note

II. **Log-Mel Spectrogram:** A logarithmically scaled Mel spectrogram that preserves more spectral information than MFCC while maintaining perceptual frequency scaling.

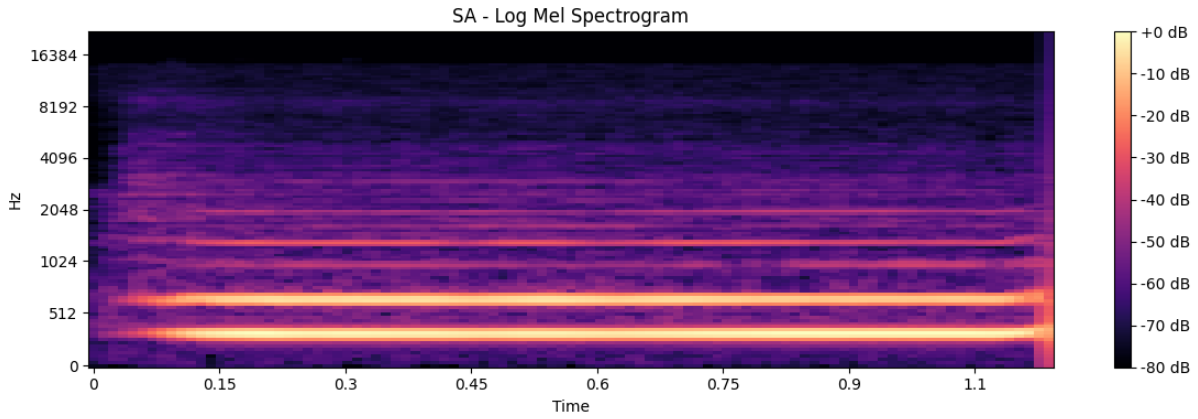


Figure 4.3: Log Mel spectrogram Visualization of an individual note

III. **Pitch:** Represents the fundamental frequency (F0) of the audio signal and provides direct information about musical note frequencies, estimated here using standard monophonic F0 estimation techniques [12, 14].

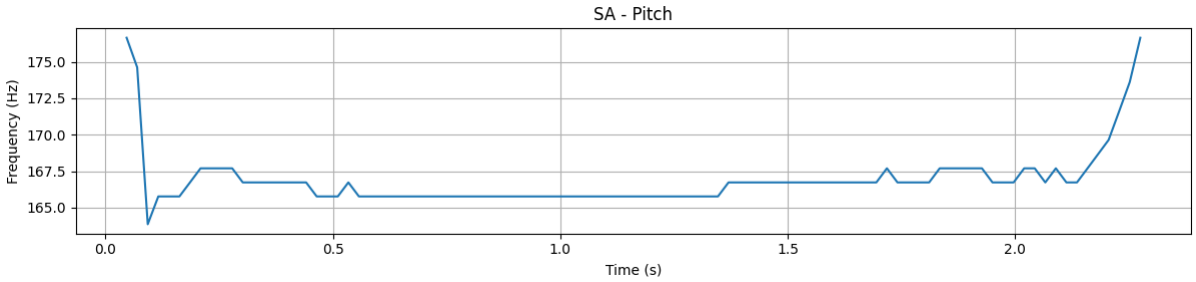


Figure 4.4: Pitch Visualization of an individual note

IV. **Root Means Square (RMS) Energy:** Measures the short-term energy of the signal and assists in detecting note onsets and offsets.

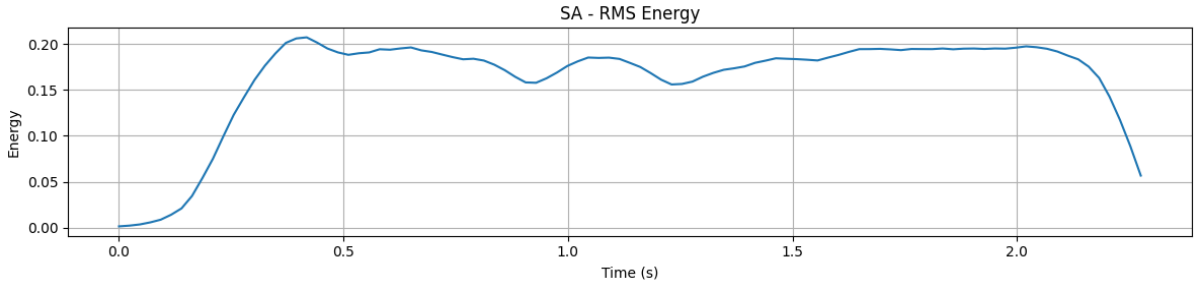


Figure 4.5: RMS energy visualization of an individual note

V. **Constant-Q Transformer (CQT):** A logarithmically spaced time-frequency representation specifically designed for musical signals [20]. Since musical notes are naturally distributed on a logarithmic frequency scale, CQT provides improved frequency resolution for note recognition and transcription tasks.

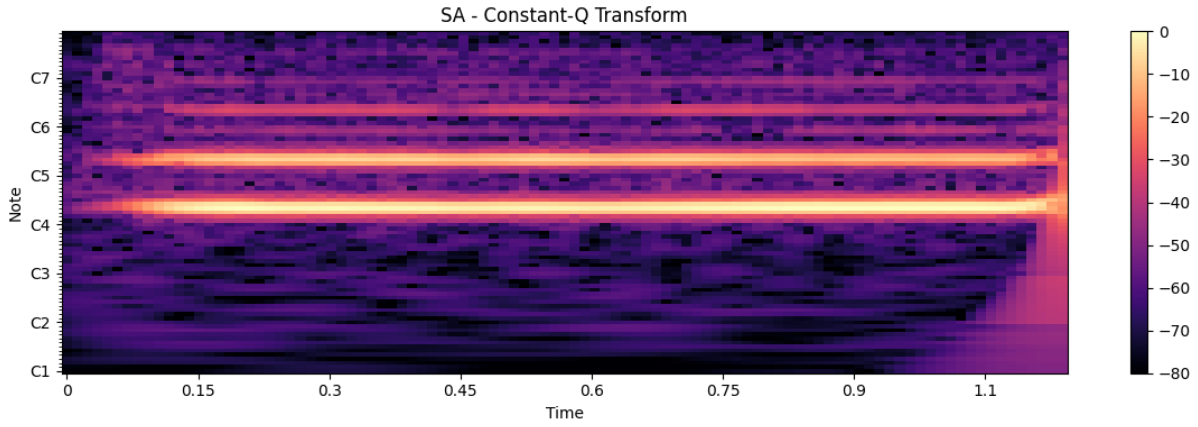


Figure 4.6: Constant-Q Transformation visualization of an individual note

Mel-frequency cepstral coefficients (MFCCs) are one of the most widely used feature representations in speech and audio classification tasks due to their compact representation of spectral characteristics. Preliminary feature extraction was performed using MFCCs during the early stages of this research. However, MFCCs provide a compressed cepstral representation that discards a significant portion of the detailed spectral information contained in the original signal.

To preserve richer time-frequency characteristics, the proposed framework ultimately employs log-Mel spectrograms as the primary spectral representation. Log-Mel spectrograms retain the two-dimensional spectral structure while remaining perceptually motivated, making them more suitable for convolutional neural networks. Additional pitch and RMS energy features were incorporated to provide complementary harmonic and energy information.

4.2.3 Experiments

To evaluate the effectiveness of different acoustic feature representations for automatic swara segmentation and classification, six experiments were conducted using the same overall processing pipeline. In all experiments, the dataset, preprocessing procedures, model architecture, training parameters, and evaluation metrics were kept identical. The only variable modified between experiments was the input feature representation. This controlled experimental design ensured that any observed differences in performance could be attributed solely to the extracted features rather than variations in the training process or model configuration.

The experimental pipeline begins with the preprocessing of the input audio, followed by feature extraction using one of the selected acoustic representations. The extracted features are then provided as input to the proposed deep learning architecture for training and inference. Finally, the trained models are evaluated using the same testing dataset and performance metrics, enabling a fair comparison of the effectiveness of each feature representation.

The following feature representations were evaluated to investigate their suitability for automatic swara segmentation and classification:

Experiment I: Log-Mel Spectrogram. Log-Mel spectrograms were selected as the baseline spectral representation because they preserve detailed time-frequency information while approximating the frequency perception of the human auditory system. Unlike compressed cepstral features, Log-Mel spectrograms retain the two-dimensional spectral structure, making them highly suitable for convolutional neural networks.

Experiment II: Pitch. Pitch information was evaluated independently to determine whether the fundamental frequency alone is sufficient for identifying individual swaras. Since each swara corresponds to a distinct musical pitch, this experiment investigates the discriminative

capability of pitch without additional spectral information.

Experiment III: RMS Energy. Root Mean Square (RMS) energy was evaluated independently to analyze the contribution of signal energy to segmentation and classification. Changes in energy often coincide with note boundaries and articulation, making RMS a potentially useful cue for detecting transitions between consecutive notes.

Experiment IV: Log-Mel Spectrogram + Pitch. This experiment combines spectral information with explicit pitch estimation. While the Log-Mel spectrogram captures the complete spectral characteristics of the audio signal, the pitch representation provides direct information about the fundamental frequency. The combination was expected to improve the discrimination between swaras with similar spectral characteristics.

Experiment V: Log-Mel Spectrogram + Pitch + RMS. The fifth experiment integrates spectral, harmonic, and energy-based information into a unified feature representation. Log-Mel spectrograms provide detailed spectral content, pitch captures melodic information, and RMS contributes temporal energy variations associated with note transitions. The objective was to determine whether these complementary features collectively improve segmentation and classification performance.

Experiment VI: Constant-Q Transform (CQT). The Constant-Q Transform was evaluated because its logarithmically spaced frequency bins closely match the musical scale, providing higher frequency resolution at lower frequencies and higher temporal resolution at higher frequencies. Since swaras correspond to discrete musical notes, CQT offers a musically meaningful representation and was expected to capture pitch-related characteristics more effectively than conventional spectral representations.

Table 4.1: Experiment Table

Experiment	**Feature Representation**	**Motivation**
Experiment 1	Log-Mel	Baseline spectral representation
Experiment 2	Pitch	Harmonic Information Only
Experiment 3	RMS	Energy information only
Experiment 4	Log-Mel + Pitch	Spectral + harmonic features
Experiment 5	Log-Mel + Pitch + RMS	Multi-feature fusion
Experiment 6	CQT	Musically aligned frequency respresentaion

4.2.4 Models

Segmentation Model. The proposed swara segmentation model is based on a Convolutional Recurrent Neural Network (CRNN) enhanced with an attention mechanism, following the same general architectural family of CNN feature extraction combined with recurrent and attention-based temporal modelling used for onset and boundary detection in prior work [25, 27, 30]. The model was designed to detect the temporal boundaries between consecutive swaras in continuous flute melodies. The convolutional layers extract local spectral patterns from the input feature representation, while the recurrent layers capture long-term temporal dependencies across the audio sequence. Finally, the attention mechanism emphasizes informative temporal regions, enabling the network to accurately identify note transition boundaries. The model was trained using the synthetic melody dataset described in Chapter 3, where frame-level ground-truth annotations were automatically generated during melody synthesis.

To ensure a fair comparison across all feature extraction experiments, the segmentation architecture and training configuration remained identical throughout Experiments I–V. The only modification between experiments was the number of input channels, corresponding to the selected feature representation. Single-feature experiments (Log-Mel, Pitch, and RMS) utilized one input channel, whereas feature fusion experiments employed two (Log-Mel + Pitch) and three (Log-Mel + Pitch + RMS) input channels, respectively. The CQT experiment followed the same segmentation framework while replacing the input representation with the Constant-Q Transform

Table 4.2: Segmentation Model Architecture

Components	**Description**
Model	CRNN (Convolutional Recurrent Neural Network)
Input	Feature tensor of size $C \times 128 \times T$
Input Channels	1 (Log-Mel / Pitch / RMS), 2 (Log-Mel + Pitch), 3 (Log-Mel + Pitch + RMS), 1 (CQT)
Convolution Block	Two Conv2D blocks with Batch Normalization [48], ReLU activation, and Max Pooling
Attention Layer	Temporal attention for emphasizing important frames
Recurrent Layers	Bidirectional LSTM Layer [49, 50], Capture temporal dependencies between consecutive frames
Output Layer	Sigmoid activation producing boundary probabilities
Objective	Detect swara transition boundaries in continuous melodies

Table 4.3: Segmentation Training Configuration

Parameter	**Value**
Dataset	Synthetic Melody Dataset
Training/Validation Split	80%/20%
Random Seed	42
Batch Size	8
Number of Epochs	25
Optimizer	Adam
Initial Learning Rate	0.001
Learning Rate Scheduler	ReduceLROnPlateau
Scheduler Patience	3 Epochs
Scheduler Reduction Factor	0.5
Loss Function	Binary Cross-Entropy with Logits Loss (BCEWithLogitsLoss)
Positive Class Weight	15
Evaluation Metrics	Precision, Recall, F1-score

Since swara boundary frames represent only a small proportion of the total audio frames, the segmentation task exhibits significant class imbalance. To address this issue, Binary Cross-Entropy with Logits Loss was employed with a positive class weight of 15, assigning greater importance to boundary frames during optimization. Since melody recordings exhibit varying durations, all feature sequences were padded to a fixed length prior to batch training. During loss computation, a binary mask was applied to exclude padded frames from contributing to the optimization process, ensuring that the model learned exclusively from valid audio samples. Model performance was evaluated using Precision, Recall, and F1-score, providing a comprehensive assessment of boundary detection accuracy.

Classification Model. The proposed swara classification model is based on a Convolutional Neural Network (CNN) enhanced with an attention mechanism. Following the segmentation stage, each isolated swara is converted into the selected feature representation and provided as input to the classifier. The convolutional layers learn discriminative spectral characteristics from the extracted features, while the attention mechanism emphasizes the most informative regions of the feature maps, enabling the network to focus on the acoustic characteristics that best distinguish individual swaras. Finally, the fully connected output layer predicts one of the seven swara classes (Sa, Re, Ga, Ma, Pa, Dha, and Ni).

To ensure a fair comparison across all feature extraction experiments, the classification architecture and training configuration remained identical throughout Experiments I–V. The only modification between experiments was the number of input channels corresponding to the selected feature representation. Single-feature experiments (Log-Mel, Pitch, and RMS) utilized one input channel, while feature fusion experiments employed two (Log-Mel + Pitch) and three (Log-Mel + Pitch + RMS) input channels, respectively. The CQT experiment followed the same classification framework while replacing the input representation with the Constant-Q

Transform.

Table 4.4: Classification Model Architecture

Component	**Description**
Model	CNN
Attention Layer	Spatial attention mechanism for emphasizing informative feature regions
Input	Feature tensor of size $C \times 64 \times 128$
Input Channels	1 (Log-Mel / Pitch / RMS), 2 (Log-Mel + Pitch), 3 (Log-Mel + Pitch + RMS), 1 (CQT)
Convolution Block	Three Conv2D blocks with Batch Normalization [48], ReLU activation, and Max Pooling
Adaptive Pooling	Adaptive Average Pooling to aggregate spectral information while preserving temporal resolution
Fully Connected Layer	Maps extracted features to swara classes
Output Layer	Fully connected layer producing probabilities for seven swara classes
Objective	Classify segmented notes into one of the seven swara classes

Table 4.5: Classification Training Configuration

Parameter	**Value**
Dataset	Individual Swara Dataset
Training/ Validation Split	80% / 20%
Random Seed	42
Batch Size	32
Number of Epochs	25
Optimizer	Adam
Initial Learning Rate	0.001
Learning Rate Scheduler	ReduceLROnPlateau
Scheduler Patience	3 Epochs
Scheduler Reduction Factor	0.5
Loss Function	Cross-Entropy Loss
Evaluation Metrics	Accuracy, Precision, Recall, F1-score

The classification model was trained using the individual swara dataset described in Chapter 3. All feature tensors were padded or truncated to a fixed temporal length to ensure uniform input dimensions during batch training. The network parameters were optimized using the Adam optimizer [51] with an initial learning rate of 0.001. A ReduceLROnPlateau scheduler was employed to automatically decrease the learning rate when the validation loss plateaued, promoting stable convergence throughout training. Model performance was evaluated using the same validation protocol across all experiments, ensuring that differences in classification performance were solely attributable to the selected feature representation.

Attention Mechanism. Integrating an attention mechanism [52] into a CRNN for audio segmentation provides a massive upgrade by allowing the network to dynamically focus on the most relevant parts of a sequence rather than treating every time frame equally. In complex audio streams—such as a continuous recording containing overlapping sounds or varying acoustic environments—different segments carry vastly different levels of importance. While recurrent layers like LSTMs process inputs sequentially, they can struggle to retain long-range context over extended audio clips. Attention mechanisms solve this by computing weighted importance scores across all time steps simultaneously, enabling the model to "attend" directly to critical transitional boundaries or specific event markers (like the exact start and end of a spoken word or environmental noise). This drastically improves temporal precision during audio segmentation.

When applied to a standalone CNN for audio classification, attention mechanisms (often implemented as spatial or channel-wise attention blocks, such as Squeeze-and-Excitation networks [53]) elevate performance by highlighting the most informative spectral regions of a Mel-spectrogram. In audio classification, a CNN extracts local texture and frequency features, but not all frequency bands or harmonic patterns contribute equally to identifying a sound class. Attention allows the model to adaptively amplify feature maps corresponding to discriminative acoustic cues—such as a specific formant in voice recognition or a distinct high-frequency snap in percussion—while suppressing background noise or irrelevant frequencies. By selectively weighting these feature channels and spatial regions, the CNN becomes significantly more robust and accurate, even when dealing with noisy or complex audio data.

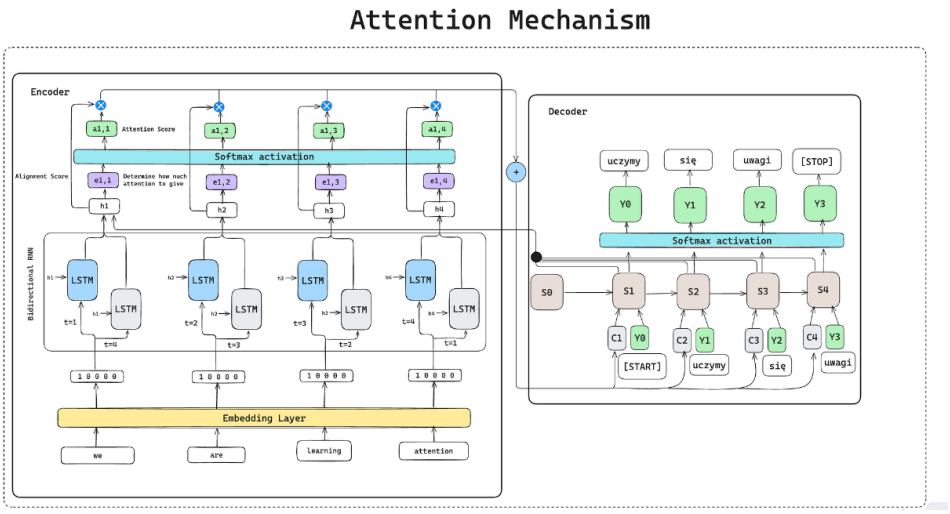


Figure 4.7: Architectural diagram of Attention Mechanism

4.2.5 Training Configurations

All experiments were implemented in Python using the PyTorch deep learning framework [54] and executed on the Google Colab platform. Model training and evaluation were accelerated using an NVIDIA Tesla T4 Graphics Processing Unit (GPU), enabling efficient processing of

both the segmentation and classification models. Identical training configurations were maintained across all experiments to ensure a fair comparison between different feature representations. The detailed software and hardware specifications used in this study are summarized in Table 4.6.

Table 4.6: Experimental Environment

Component	**Specification**
Programming Language	Python 3
Deep Learning Framework	PyTorch
Audio Processing Library	Librosa [55]
Numerical Computing	NumPy
Data Manipulation	Pandas
Visualization	Matplotlib
Development Environment	Google Colab
Hardware Accelerator	NVIDIA Tesla T4 GPU
Operating Environment	Ubuntu(Google Colab Runtime)

4.2.6 Evaluation Metrics

The performance of the proposed framework was evaluated using several widely adopted classification and segmentation metrics, including Accuracy, Precision, Recall, and F1-score. These metrics provide complementary insights into the effectiveness of the proposed models by measuring both the correctness of predictions and the ability to accurately detect swara boundaries.

Accuracy: It measures the overall proportion of correctly predicted samples among all predictions and is defined as

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{4.1}$$

where TP, TN, FP, and FN denote True Positives, True Negatives, False Positives, and False Negatives, respectively.

Precision: It evaluates the proportion of predicted positive samples that are actually correct. It measures how reliable the positive predictions of the model are and is expressed as

$$\text{Precision} = \frac{TP}{TP + FP} \tag{4.2}$$

Recall: It measures the ability of the model to correctly identify all positive samples and is calculated as

$$\text{Recall} = \frac{TP}{TP + FN} \tag{4.3}$$

The **F1-score** is the harmonic mean of Precision and Recall, providing a balanced evaluation when both false positives and false negatives are important. It is computed as

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{4.4}$$

The F1-score is particularly suitable for imbalanced datasets, where relying solely on Accuracy may produce misleading conclusions.

Evaluation Protocol

Different evaluation metrics were employed for the segmentation and classification tasks according to their respective objectives.

For the classification model, Accuracy and F1-score were selected as the primary evaluation metrics. Accuracy measures the overall correctness of swara prediction, while the F1-score provides a balanced assessment of classification performance by considering both Precision and Recall, making it more suitable when class distributions are not perfectly uniform.

For the segmentation model, Precision, Recall, and F1-score were used to evaluate the detection of swara boundaries. Precision measures how many predicted boundaries correspond to actual swara transitions, Recall measures how many true boundaries are successfully detected, and the F1-score provides a balanced measure of segmentation performance by combining both metrics.

Chapter 5

Result and Discussion

To evaluate the effectiveness of the proposed framework, six experiments were conducted using different acoustic feature representations while maintaining identical model architectures and training configurations. This ensured that any variation in performance could be attributed solely to the selected feature representation rather than differences in model design or optimization.

Based on the literature reviewed in Chapter 2, Log-Mel spectrograms have been widely adopted for audio and music processing tasks because of their ability to represent both spectral and perceptual characteristics of sound. Therefore, it was initially hypothesized that Log-Mel would achieve the best overall performance for both swara segmentation and classification. In addition to Log-Mel, Pitch, RMS, Log-Mel + Pitch, Log-Mel + Pitch + RMS, and Constant-Q Transform (CQT) representations were investigated. While all feature representations produced satisfactory results to varying degrees, the CQT-based model achieved the highest overall performance across the proposed end-to-end framework.

5.1 Experiment 1: Log-Mel

The first experiment utilized the Log-Mel spectrogram as the input feature representation for both the segmentation and classification models. Based on previous studies in music information retrieval and audio classification, Log-Mel was initially expected to provide the highest overall performance due to its ability to capture both spectral and perceptual characteristics of audio signals.

For swara classification, the CNN with Attention model was trained for 30 epochs using the Log-Mel feature representation. The model consisted of 101,960 trainable parameters and achieved an overall classification accuracy of 85.0% with an F1-score of 89.3%. The training and validation loss and accuracy curves are presented in Figure 5.1. The smooth convergence of both training and validation curves indicates stable learning without significant overfitting, demonstrating that Log-Mel provides sufficient spectral information for accurate swara classification.

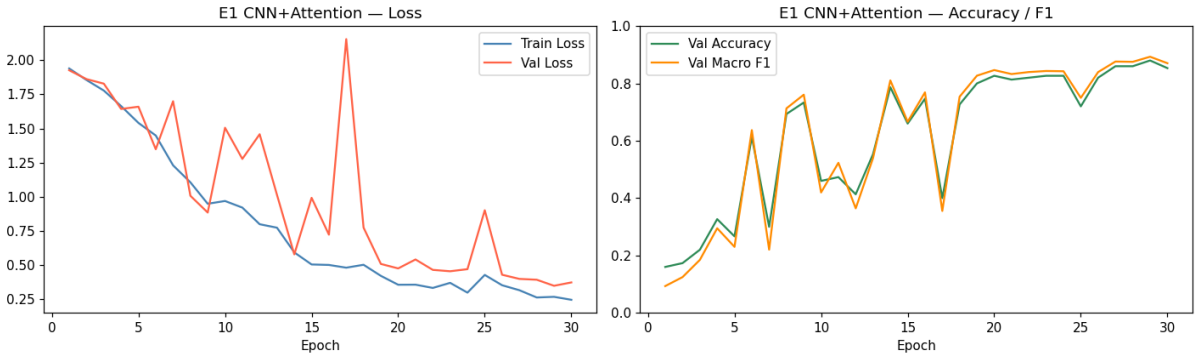


Figure 5.1: Training and validation loss, accuracy, and F1-score of the classification model for Experiment 1.

For swara segmentation, the CRNN with Attention model was trained for 25 epochs using the same Log-Mel feature representation. The segmentation network contained 1,090,594 trainable parameters. The model was evaluated using Precision, Recall, and F1-score for boundary detection. The corresponding training curves are illustrated in Figure 5.2, showing the progression of the segmentation model throughout training.

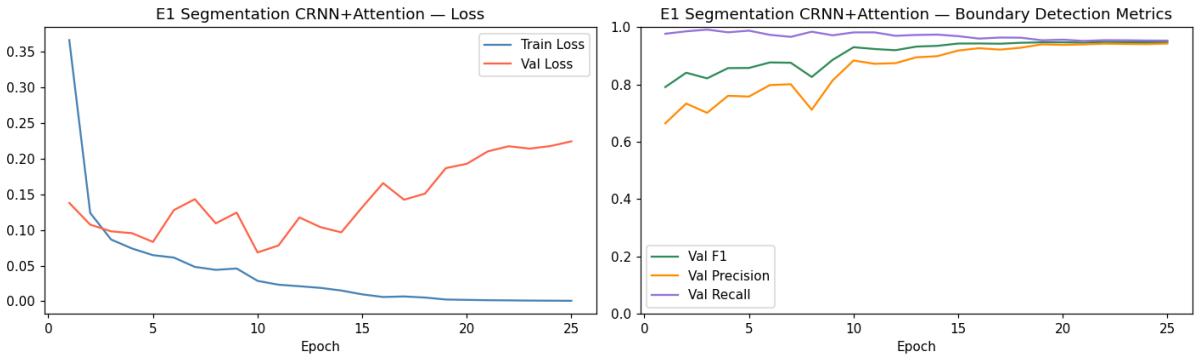


Figure 5.2: Training loss and boundary detection metrics (Precision, Recall, and F1-score) of the segmentation model for Experiment 1.

5.1.1 End to end pipeline for experiment 1

After independently training the segmentation and classification models, both components were integrated to evaluate the complete swara identification pipeline. Every test melody was processed through the segmentation stage followed by the classification stage using the trained Log-Mel models.

Segmentation performance was measured based on the percentage of melodies that were segmented perfectly, where every swara boundary was correctly identified. For classification evaluation, only these perfectly segmented melodies were considered to ensure that the reported classification accuracy reflected the capability of the classifier rather than segmentation errors. Using this evaluation protocol, the proposed framework achieved a segmentation accuracy of

71.43%, while the classification stage achieved a note-level accuracy of 84.29% on the correctly segmented melodies. An example of the complete transcription generated by the proposed pipeline is shown in Figure 5.3.

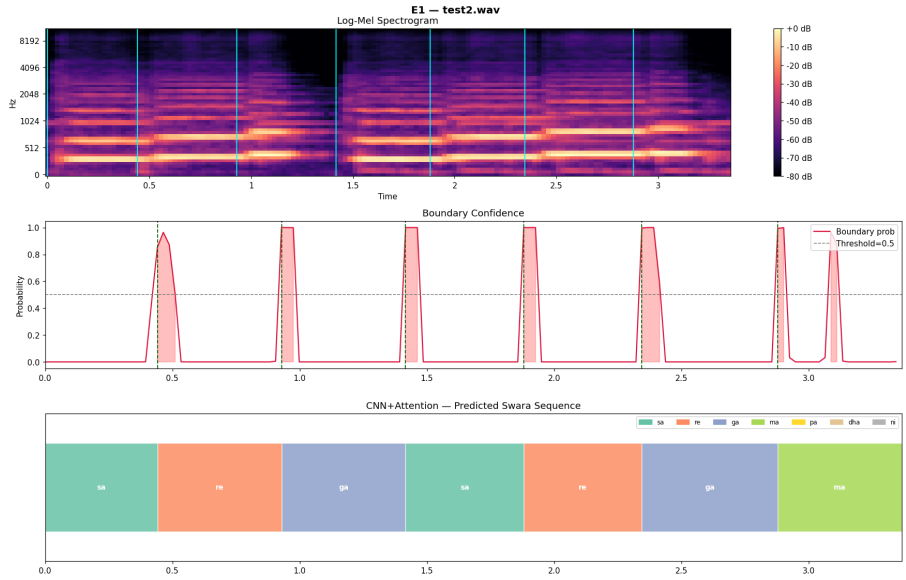


Figure 5.3: End to end pipeline visualization of experiment 1

5.2 Experiment 2: Pitch

The second experiment investigated the effectiveness of Pitch as the sole feature representation for both segmentation and classification. Unlike Log-Mel, which contains rich spectral information, Pitch represents only the fundamental frequency of the audio signal. This experiment was conducted to evaluate whether pitch information alone is sufficient for accurate swara identification.

The CNN with Attention classifier was trained using the same architecture and training configuration as Experiment 1, with Pitch replacing Log-Mel as the input feature. The model achieved an overall classification accuracy of 69.0% and an F1-score of 71.5%. Compared with Experiment 1, a noticeable decrease in performance was observed, indicating that although Pitch effectively captures note frequency, it lacks the harmonic and timbral information necessary for reliable discrimination between acoustically similar swaras.

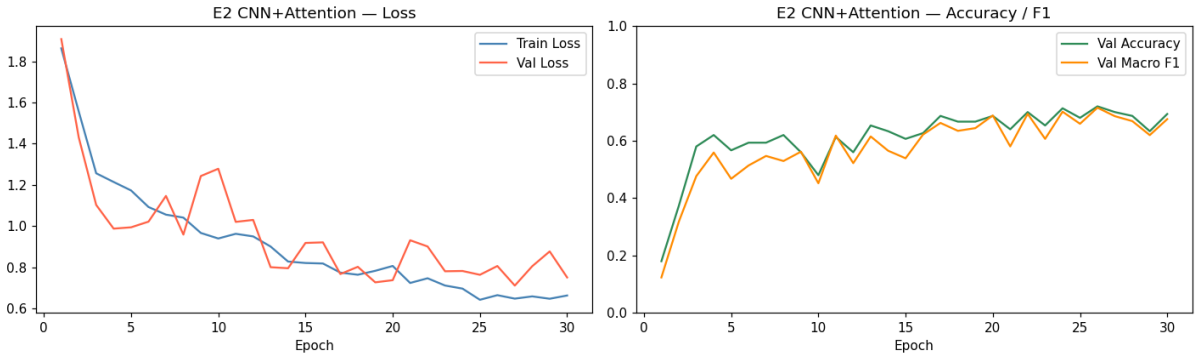


Figure 5.4: Training and validation loss, accuracy, and F1-score of the classification model for Experiment 2.

For segmentation, the CRNN with Attention model followed the identical architecture and training configuration used in Experiment 1. The model achieved a Precision of 47.0%, Recall of 93.4%, and an F1-score of 57.7%. The high Recall indicates that the model successfully detected most true swara boundaries; however, the comparatively low Precision suggests that a considerable number of false boundary predictions were produced, resulting in a lower overall F1-score.

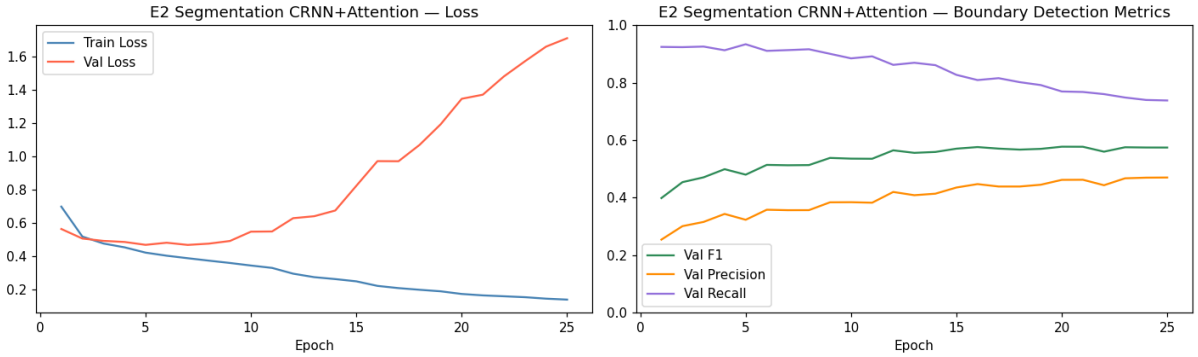


Figure 5.5: Training loss and boundary detection metrics (Precision, Recall, and F1-score) of the segmentation model for Experiment 2.

5.2.1 End to End Pipeline for Experiment 2

The complete end-to-end pipeline was evaluated following the same procedure as Experiment 1. Every test melody was processed through the segmentation and classification stages using Pitch as the input feature representation. Based on the defined evaluation protocol, the proposed framework achieved a segmentation accuracy of 40.00%. Among the perfectly segmented melodies, the classification model achieved a note-level accuracy of 21.43%. The significant reduction in end-to-end performance compared with Experiment 1 demonstrates that Pitch alone is insufficient for building a robust swara identification system, particularly for segmentation where reliable spectral information is essential for accurate boundary detection.

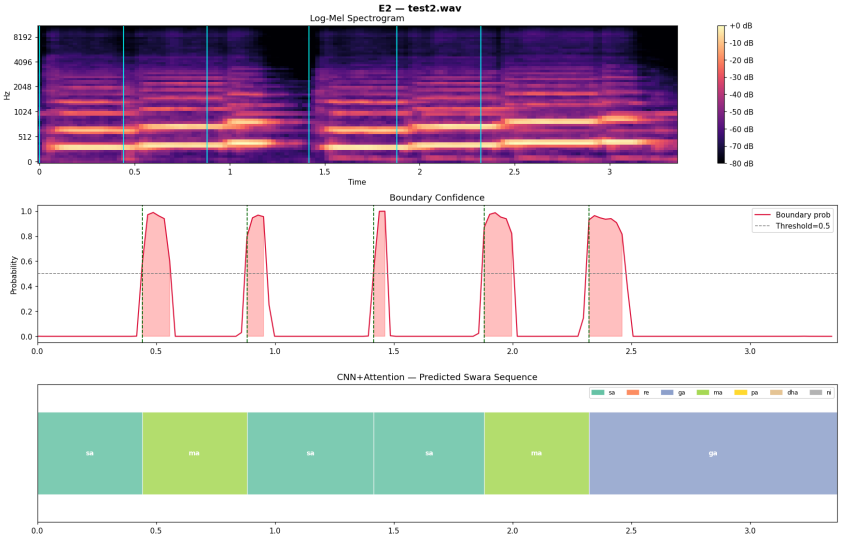


Figure 5.6: End to end pipeline visualization of experiment 2

5.3 Experiment 3: RMS

The third experiment investigated the effectiveness of Root Mean Square (RMS) Energy as the sole feature representation for both swara segmentation and classification. Unlike Log-Mel and Pitch, RMS represents only the energy or loudness variations of the audio signal and does not contain any spectral or frequency information. This experiment was conducted to evaluate whether energy alone could provide sufficient information for identifying swaras.

For swara classification, the CNN with Attention model was trained using the RMS feature representation while maintaining the same architecture and training configuration as the previous experiments. The model achieved an overall classification accuracy of 31.0% with an F1-score of 29.6%. The considerably lower performance indicates that RMS energy alone is insufficient for distinguishing between different swaras, as multiple notes may exhibit similar energy patterns despite having different spectral characteristics.

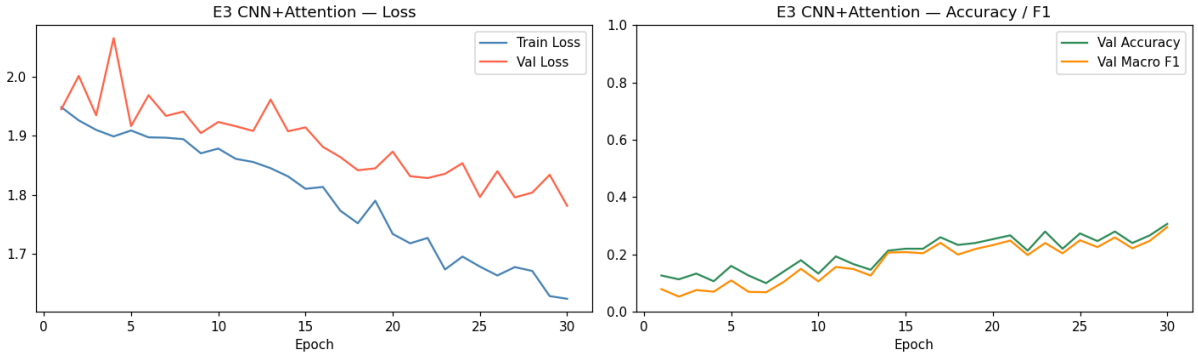


Figure 5.7: Training and validation loss, accuracy, and F1-score of the classification model for Experiment 3.

For swara segmentation, the CRNN with Attention model achieved a Precision of 61.9%, Recall of 97.2%, and an F1-score of 74.8%. Although RMS lacks frequency information, the model was still able to identify a large proportion of swara boundaries, as reflected by the high Recall. However, the comparatively lower Precision suggests that the model generated a considerable number of false boundary detections.

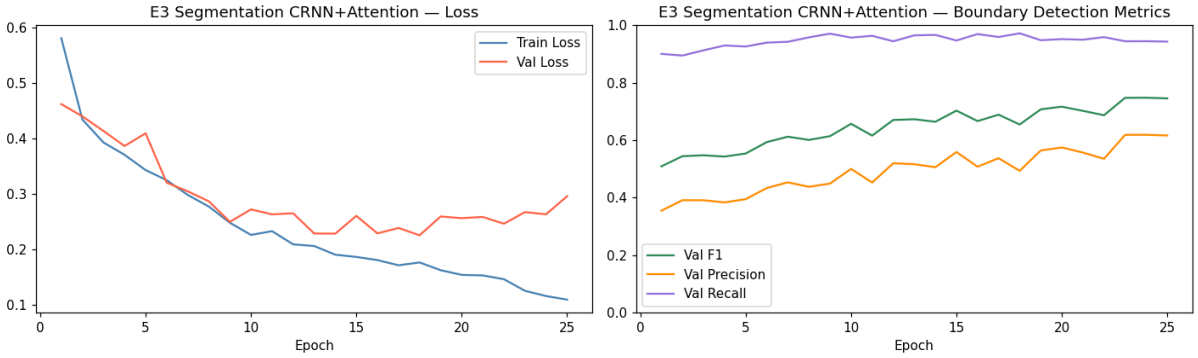


Figure 5.8: Training loss and boundary detection metrics (Precision, Recall, and F1-score) of the segmentation model for Experiment 3.

5.3.1 End-to-End Pipeline

The complete end-to-end pipeline was evaluated following the same procedure used in the previous experiments. Every test melody was processed through both the segmentation and classification stages using RMS as the input feature representation. The proposed framework achieved a segmentation accuracy of 64.29%. Among the perfectly segmented melodies, the classification model achieved a note-level accuracy of 14.29%. Although RMS provided moderate segmentation performance, the lack of discriminative spectral information resulted in poor classification performance, making it unsuitable as a standalone feature for automatic swara identification.

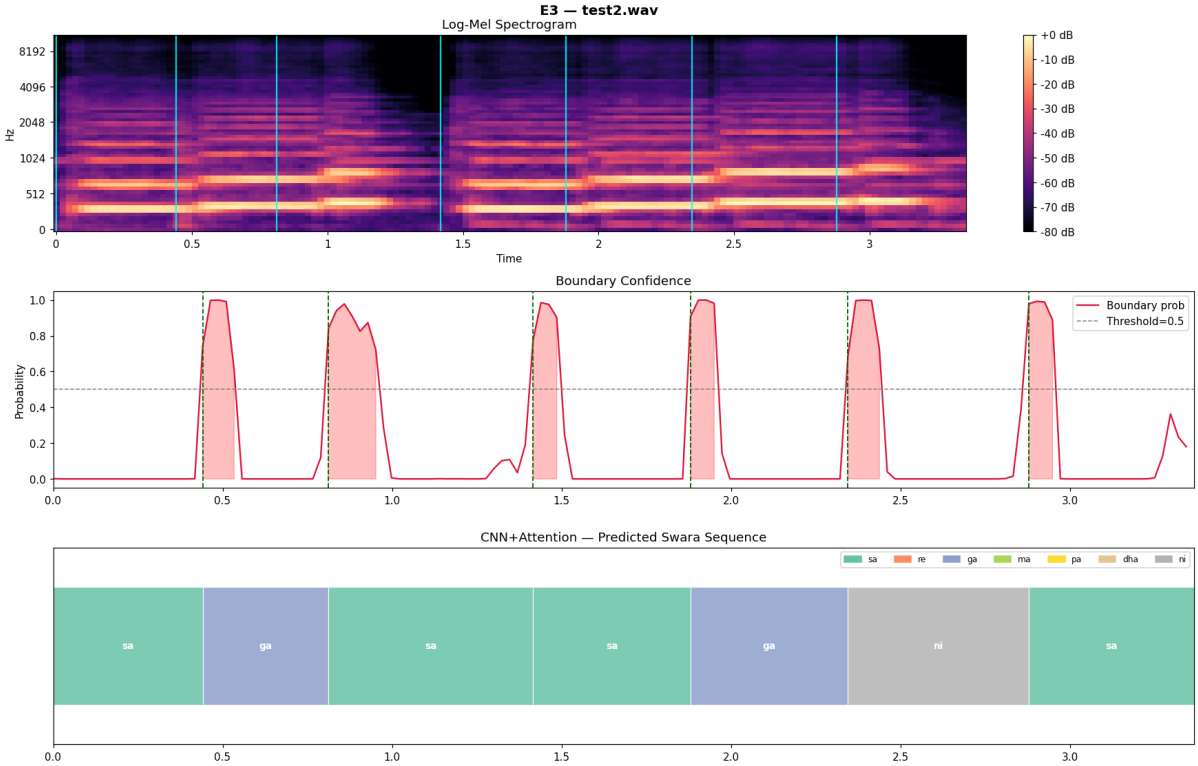


Figure 5.9: Example visualization of the end-to-end swara segmentation and classification pipeline for Experiment 3.

5.4 Experiment 4: Log-Mel + Pitch

The fourth experiment combined Log-Mel spectrograms and Pitch to investigate whether combining spectral and frequency information could improve segmentation and classification performance. The same model architectures and training configurations were maintained, with the only difference being the use of two input channels corresponding to the fused feature representation.

For swara classification, the CNN with Attention model achieved an overall classification accuracy of 84.7% with an F1-score of 86.5%. The combined representation produced performance comparable to the Log-Mel-only model, indicating that incorporating Pitch information provided only a marginal improvement in classification performance.

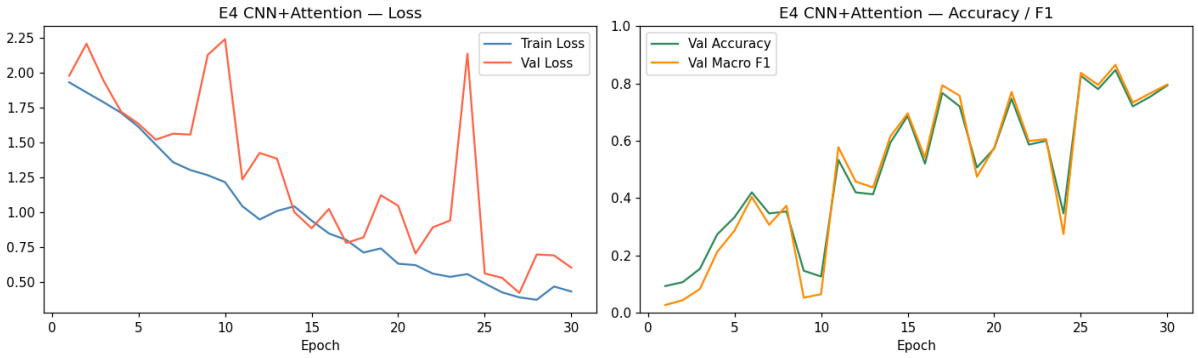


Figure 5.10: Training and validation loss, accuracy, and F1-score of the classification model for Experiment 4.

For swara segmentation, the CRNN with Attention model achieved a Precision of 94.3%, Recall of 98.7%, and an F1-score of 94.9%. These results represent a substantial improvement over the previous experiments, demonstrating that the combination of spectral and pitch information enables the model to identify swara transition boundaries with significantly higher reliability.

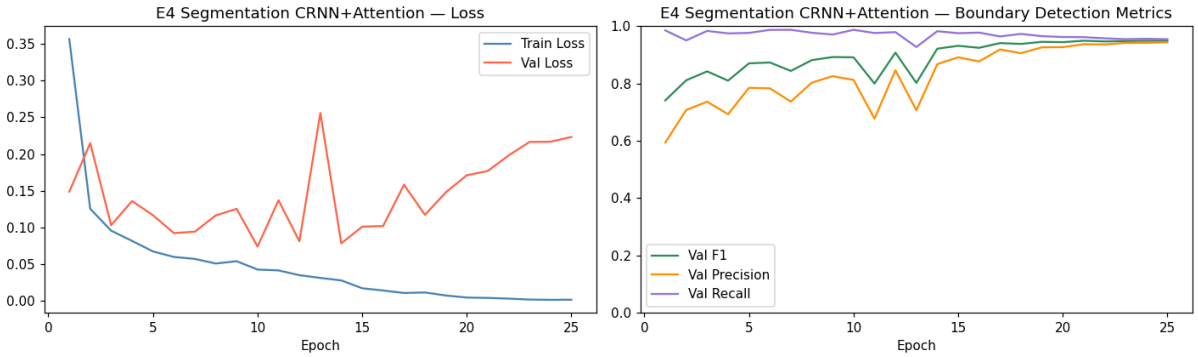


Figure 5.11: Training loss and boundary detection metrics (Precision, Recall, and F1-score) of the segmentation model for Experiment 4.

5.4.1 End-to-End Pipeline Evaluation

The complete pipeline was evaluated using the combined Log-Mel and Pitch representation. The proposed framework achieved a segmentation accuracy of 57.14%, while the classification stage achieved a note-level accuracy of 39.29% on the correctly segmented melodies. Although the segmentation model achieved excellent frame-level metrics during validation, the end-to-end evaluation demonstrates that perfectly segmenting an entire melody remains a challenging task, resulting in lower overall pipeline performance.

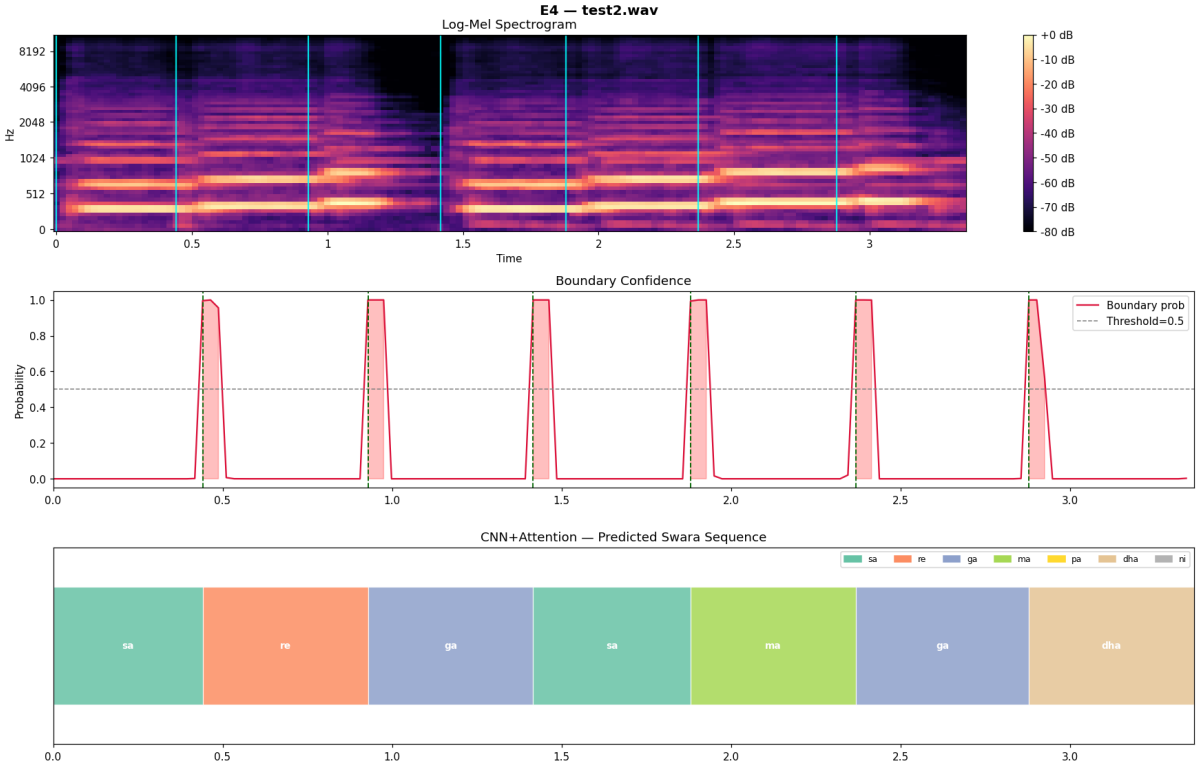


Figure 5.12: Example visualization of the end-to-end swara segmentation and classification pipeline for Experiment 4.

5.5 Experiment 5: Log-Mel + Pitch + RMS

The fifth experiment investigated the combined use of Log-Mel, Pitch, and RMS as a three-channel input representation. This experiment aimed to determine whether incorporating energy information alongside spectral and pitch features could further improve the performance of the proposed framework.

For swara classification, the CNN with Attention model achieved an overall classification accuracy of 93.3% with an F1-score of 94.0%, representing a significant improvement over the previous experiments. The combination of complementary acoustic information enabled the classifier to learn more discriminative feature representations, resulting in the highest classification performance observed thus far.

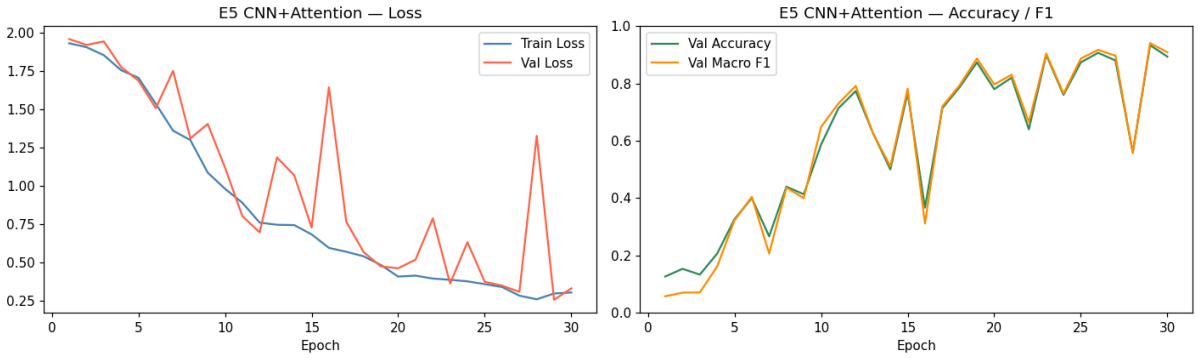


Figure 5.13: Training and validation loss, accuracy, and F1-score of the classification model for Experiment 5.

For swara segmentation, the CRNN with Attention model achieved a Precision of 93.5%, Recall of 99.3%, and an F1-score of 94.4%. Similar to Experiment 4, the segmentation model demonstrated excellent boundary detection performance, with both Precision and Recall remaining consistently high throughout training.

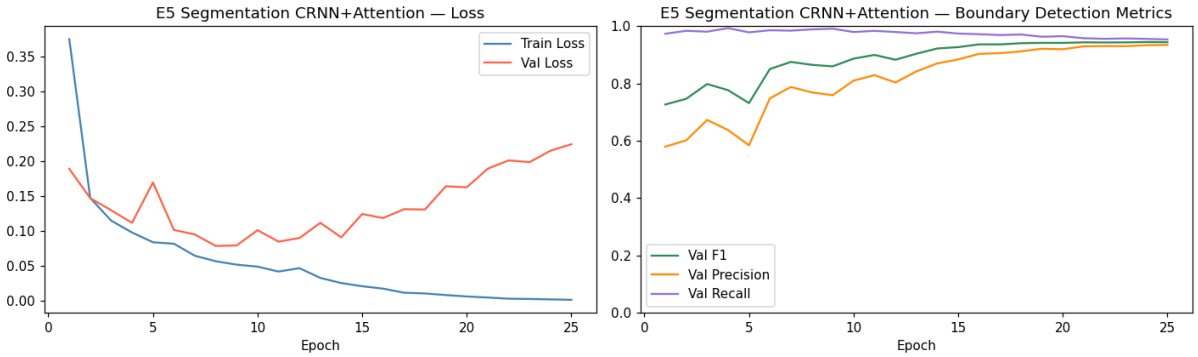


Figure 5.14: Training loss and boundary detection metrics (Precision, Recall, and F1-score) of the segmentation model for Experiment 5.

5.5.1 End-to-End Pipeline Evaluation

The complete end-to-end pipeline was evaluated using the three-channel feature representation. The proposed framework achieved a segmentation accuracy of 50.00%. Among the perfectly segmented melodies, the classification model achieved a note-level accuracy of 75.51%. The substantial improvement in classification accuracy demonstrates the effectiveness of combining complementary acoustic features. However, the lower melody-level segmentation accuracy indicates that accurate boundary detection across an entire melody remains the primary factor limiting the overall performance of the complete pipeline.

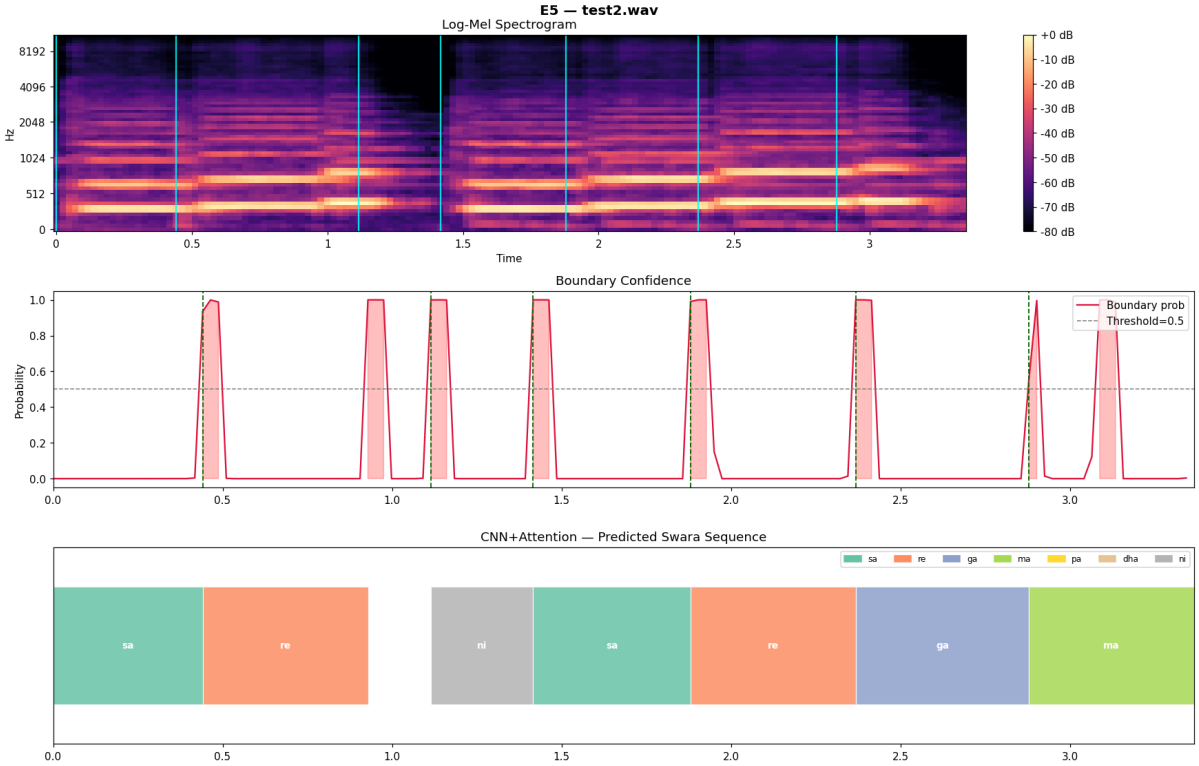


Figure 5.15: Example visualization of the end-to-end swara segmentation and classification pipeline for Experiment 5.

5.6 Experiment 6: Constant-Q Transform (CQT)

The sixth experiment utilized the Constant-Q Transform (CQT) as the input feature representation for both the segmentation and classification models. Unlike the previous feature representations, CQT employs logarithmically spaced frequency bins that closely correspond to the musical scale, making it particularly suitable for music signal analysis and monophonic instrument recognition. Since flute performances consist of discrete musical notes with well-defined harmonic structures, CQT was expected to provide a more informative representation than conventional spectrogram-based features.

For swara classification, the CNN with Attention model was trained for 30 epochs using the same network architecture and training configuration employed in the previous experiments. Throughout training, both the training and validation accuracies increased consistently, indicating stable convergence of the model. The best validation accuracy of 99.33% was achieved at Epoch 26, after which the performance remained stable until the completion of training. These results demonstrate that the CQT representation provides highly discriminative information for distinguishing between the seven swara classes.

For swara segmentation, the CRNN with Attention model was trained for 20 epochs using the CQT feature representation. The training loss decreased steadily from 0.0704 during the first

epoch to 0.0257 at the final epoch, indicating stable optimization and convergence throughout the training process. The consistently decreasing loss suggests that the model effectively learned the temporal characteristics required to identify swara transition boundaries within continuous flute melodies.

5.6.1 End-to-End Pipeline Evaluation

The complete end-to-end pipeline was evaluated following the same procedure adopted for the previous experiments. Every test melody was processed through both the segmentation and classification stages using the trained CQT models. Segmentation performance was measured based on the percentage of melodies that were segmented perfectly, while classification accuracy was calculated only for the correctly segmented melodies.

The proposed framework achieved the highest overall performance among all six experiments, obtaining a segmentation accuracy of 92.86%. Furthermore, the classification model correctly identified 97.80% of the swaras within the perfectly segmented melodies. These results demonstrate that the Constant-Q Transform provides the most suitable acoustic representation for the proposed end-to-end swara segmentation and classification framework, outperforming all other investigated feature representations.

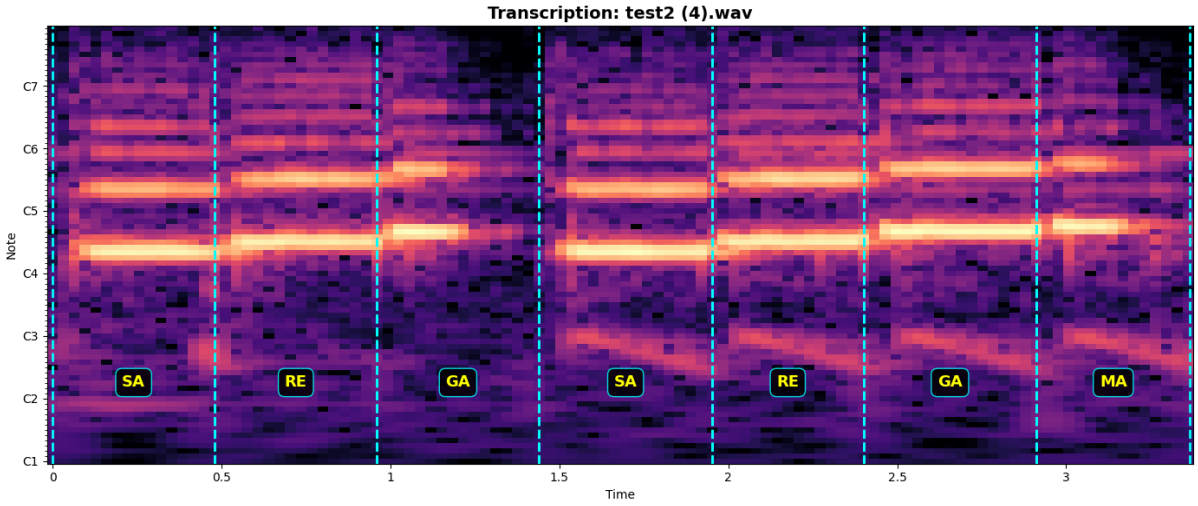


Figure 5.16: Example visualization of the end-to-end swara segmentation and classification pipeline for Experiment 6.

Table 5.1: Overall Performance Comparison of All Experiment

Experiment	Feature	Classification Accuracy (%)	Classification F1 (%)	Segmentation Precision (%)	Segmentation Recall	Segmentation F1	End-to end Segmentation	End to end Classification
E1	Log-Mel	85.00	89.30	94.3	98.9	95.0	71.43	84.29
E2	Pitch	69.00	71.50	47.00	93.40	57.70	40.00	21.43
E3	RMS	31.00	29.60	61.90	97.20	74.80	64.29	14.29
E4	Log-Mel + Pitch	84.70	86.50	94.30	98.70	94.90	57.14	39.29
E5	Log-Mel + Pitch + RMS	93.30	94.00	93.50	99.30	94.40	50.00	75.51
E6	CQT	99.33					92.86	97.80

5.7 Discussion

The experimental results demonstrate that the choice of acoustic feature representation has a significant impact on both swara segmentation and classification performance. Although all experiments employed identical model architectures, training procedures, optimization strategies, and evaluation protocols, considerable performance differences were observed across the six investigated feature representations. These variations indicate that the quality and nature of the extracted acoustic features play a more influential role than the model architecture itself.

Among the six experiments, the Constant-Q Transform (CQT) consistently achieved the best overall performance. The end-to-end pipeline attained a segmentation accuracy of 92.86% and a note-level classification accuracy of 97.80%, outperforming every other feature representation evaluated in this study. Unlike conventional spectrogram-based features, CQT represents frequency on a logarithmic scale that closely matches the spacing of musical notes. Since Indian classical flute performances are inherently musical and consist of discrete swaras, this representation preserves harmonic relationships more effectively, enabling both the segmentation and classification models to learn more discriminative musical patterns.

The Log-Mel spectrogram also demonstrated strong performance throughout the experiments. As one of the most widely used feature representations in speech and music processing, Log-Mel effectively captures both spectral and perceptual characteristics of audio signals. The experimental results confirm that Log-Mel provides a reliable representation for automatic swara identification, producing satisfactory segmentation and classification performance. However, despite its strong results, it remained inferior to the CQT representation, suggesting that feature representations specifically designed for musical signals are more suitable for this application.

Experiments utilizing Pitch and RMS individually produced considerably lower performance. Pitch represents only the fundamental frequency of an audio signal while ignoring harmonic content and timbral information, making it difficult for the models to distinguish between acoustically similar swaras. RMS performed even worse for classification because energy information alone contains little discriminative information regarding musical note identity. Interestingly, although RMS achieved relatively high Recall during segmentation, its Precision remained comparatively low, indicating that the model detected many true note boundaries while simultaneously producing a large number of false boundary predictions. These observations demonstrate that neither Pitch nor RMS alone provides sufficient information for robust end-to-end swara identification.

The feature fusion experiments further illustrate the complementary nature of different acoustic representations. Combining Log-Mel with Pitch substantially improved the segmentation metrics, indicating that spectral and frequency information together provide a richer representation of note transitions. Furthermore, incorporating RMS into the feature set resulted in the highest classification performance among the non-CQT experiments. This suggests that while RMS alone is insufficient for accurate classification, it contributes useful complementary in-

formation when combined with richer spectral features. The results demonstrate that feature fusion enables the models to exploit multiple acoustic characteristics simultaneously, leading to improved overall performance.

An interesting observation can be made by comparing the segmentation evaluation metrics with the end-to-end segmentation accuracy. Several experiments achieved frame-level F1-scores exceeding 94%, yet their melody-level segmentation accuracies remained considerably lower. This difference arises from the evaluation protocol adopted in this study. Frame-level Precision, Recall, and F1-score evaluate the correctness of individual boundary predictions across all frames, whereas the end-to-end segmentation metric considers a melody to be correctly segmented only when every swara boundary within that melody is identified accurately. Consequently, even a single missed or falsely detected boundary causes the entire melody to be counted as incorrectly segmented. This evaluation criterion is considerably more stringent and better reflects the practical performance of a complete automatic transcription system.

Overall, the experimental results validate the effectiveness of the proposed end-to-end framework for automatic swara segmentation and classification. More importantly, they demonstrate that selecting an appropriate feature representation is essential for achieving robust performance. Among all investigated features, the Constant-Q Transform proved to be the most effective representation for monophonic flute melodies, enabling the proposed framework to achieve the highest segmentation and classification performance while maintaining a fully automated pipeline requiring no manually annotated continuous recordings.

Chapter 6

Limitations

Although the proposed framework achieved promising results for automatic swara segmentation and classification, several limitations remain.

First, the experiments were conducted exclusively using flute recordings. While the proposed synthetic data generation strategy and end-to-end pipeline were designed for monophonic musical instruments in general, the present study only validates the framework on flute audio. Consequently, although the methodology is expected to be applicable to other monophonic instruments, such as the violin, recorder, clarinet, or saxophone, additional experiments are required to verify its generalization capability.

Second, the proposed system considers only the seven fundamental swaras (Sa, Re, Ga, Ma, Pa, Dha, and Ni). Expressive musical ornamentations commonly found in Indian classical music, including Meend, Gamak, Murki, and Kan Swara, were not incorporated into the current dataset or evaluation, and prior work has shown that such ornamentation and melodic context substantially shape the acoustic realization of a note [41, 42]. As a result, the framework has not yet been assessed on more complex performance techniques.

Third, although the synthetic melody generation process introduced variations through multiple transition styles and digital signal processing augmentations, synthetically generated melodies cannot perfectly reproduce every expressive characteristic of real-world performances. Variations in performer style, breathing techniques, articulation, and recording environments may introduce acoustic characteristics that are not fully represented in the generated dataset.

Finally, the proposed framework assumes monophonic audio, where only a single note is present at any given time. Therefore, the current approach is not directly applicable to polyphonic music or recordings containing multiple simultaneous instruments without significant modifications to both the segmentation and classification stages.

Chapter 7

Future Work

The proposed framework provides a foundation for automatic swara segmentation and classification using synthetically generated melodies. Although the obtained results are promising, several directions exist for extending and improving the proposed system.

The most immediate future work is to evaluate the proposed pipeline on other monophonic musical instruments. While this study focused exclusively on flute recordings, the underlying methodology is independent of any instrument-specific characteristics. By replacing the isolated note recordings with those of other monophonic instruments, such as the violin, recorder, clarinet, saxophone, or similar instruments, the same synthetic melody generation pipeline can be employed to create automatically annotated datasets. Experimental validation across multiple instruments would provide stronger evidence regarding the generalization capability of the proposed framework.

Another important extension is the inclusion of more expressive musical techniques commonly observed in Indian classical music. The current work considers only the seven fundamental swaras, whereas practical performances frequently contain ornamentations such as Meend, Gamak, Murki, and Kan Swara. Incorporating these expressive elements into the synthetic melody generation process would enable the proposed framework to operate on more realistic and musically complex performances.

Although the proposed synthetic dataset successfully reduced the need for manually annotated continuous recordings, future work may further enhance the realism of the generated melodies. More sophisticated transition modeling, performer-specific playing styles, breathing effects, expressive dynamics, and additional acoustic augmentations could be incorporated to better simulate professional performances. Such improvements would further narrow the gap between synthetic and real-world musical recordings.

Since the synthetic melody generation process is fully automated, future research could investigate generating substantially larger datasets with thousands or even millions of automatically annotated melodies. Such large-scale synthetic datasets may facilitate the training of more complex deep learning models while eliminating the need for extensive manual annotation.

Future research may also investigate more advanced deep learning architectures. While this work employed CNN with Attention and CRNN with Attention models, recent sequence modeling techniques such as Transformer-based architectures [52], Conformers [56], and self-supervised audio representation models [57] could potentially improve both segmentation and classifica-

tion performance, particularly for longer and more complex musical recordings.

Finally, the proposed framework can be extended toward a complete real-time automatic music transcription system. Integrating segmentation, swara classification, notation generation, and visualization into a single application would enable musicians, educators, and researchers to analyze monophonic performances automatically. Such a system could support music education, digital archiving of traditional performances, and future music information retrieval applications.

Chapter 8

Conclusion

Automatic swara segmentation and classification play an important role in music information retrieval, digital music analysis, and computer-assisted music education. However, developing robust automatic transcription systems remains challenging due to the scarcity of accurately annotated continuous musical recordings. Manual annotation of note boundaries is a labor-intensive and time-consuming process, making it difficult to construct large datasets for training deep learning models. Motivated by this challenge, this research proposed an end-to-end framework that minimizes the dependence on manually annotated continuous recordings by utilizing synthetically generated melodies with automatically created ground-truth annotations.

The proposed framework consists of four major stages: dataset preparation, feature extraction, swara segmentation, and swara classification. An individual swara dataset containing recordings of the seven fundamental swaras was first collected from both publicly available recordings and self-recorded flute performances. These isolated recordings were then used to generate a synthetic melody dataset through randomized note selection, transition modeling, and multiple digital signal processing augmentations. Since the melody generation process automatically recorded the temporal boundaries of every inserted note, accurate frame-level annotations were obtained without requiring any manual labeling.

To evaluate the proposed framework, six different acoustic feature representations were investigated, namely Log-Mel, Pitch, RMS, Log-Mel + Pitch, Log-Mel + Pitch + RMS, and Constant-Q Transform (CQT). All experiments employed identical segmentation and classification architectures, ensuring that performance differences resulted solely from the selected feature representation. Experimental results demonstrated that the Constant-Q Transform provided the most effective representation for monophonic flute melodies, enabling the proposed framework to achieve the highest overall performance, with an end-to-end segmentation accuracy of 92.86% and a note-level classification accuracy of 97.80% on correctly segmented melodies.

Beyond the achieved performance, the primary contribution of this research lies in demonstrating that accurate swara segmentation and classification can be accomplished without relying on hundreds of hours of manually annotated continuous recordings. The proposed synthetic melody generation strategy significantly reduces the annotation effort while preserving reliable ground-truth information for model training and evaluation. This provides a practical and scalable alternative for developing automatic music transcription systems where annotated datasets are limited or unavailable.

Although the present work was validated only on flute recordings, the proposed methodology

is not inherently restricted to a single instrument. Since the pipeline operates on isolated recordings of individual notes, the same synthetic data generation strategy can potentially be adapted to other monophonic musical instruments by replacing the source recordings while preserving the remaining framework. This flexibility provides a promising foundation for future research in automatic music transcription and music information retrieval.

In conclusion, this thesis demonstrates that combining synthetic dataset generation with deep learning-based segmentation and classification models offers an effective solution for automatic swara identification. The proposed framework not only achieves promising performance but also presents a practical approach for reducing manual annotation requirements, making it a valuable contribution toward scalable and data-efficient research in computational music analysis.

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